

Tight Constants in Talagrand's Boolean Convolution Inequality

Zhao Song*

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Abstract

We study the optimal dimension-free constant $C_*(a)$ in Talagrand's Boolean convolution inequality $\Pr_\mu[T_a f \geq u] \leq C_*(a)/(u\sqrt{\log u})$, where T_a is the noise operator, μ is the uniform measure on the Boolean cube, $f \geq 0$ satisfies $\mathbb{E}_\mu[f] = 1$, and $u > 1$. We establish explicit upper bounds for every $0 < a < 1$ and prove $\limsup_{a \uparrow 1} C_*(a)/\sqrt{\kappa_a} < 0.3577$, where $\kappa_a := (1+a)/(1-a)$. We establish the lower bound $\liminf_{a \uparrow 1} C_*(a)/\sqrt{\kappa_a} \geq 0.3570$.

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*magic.linuxkde@gmail.com. The author would like to thank Lichen Zhang for very helpful discussions.

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1 Introduction

On the Boolean cube, Markov’s inequality gives the dimension-free estimate $\Pr_\mu[T_a f \geq u] \leq 1/u$ when $\mathbb{E}_\mu[f] = 1$, where T_a is the noise operator with correlation parameter $0 < a < 1$. Talagrand’s convolution conjecture asks for the additional factor $1/\sqrt{\log u}$, with a constant depending only on a and not on the dimension or the input density [Tal89]. This formulation measures smoothing at the level of tail probabilities even for highly concentrated inputs, for which no uniform bound on the maximum value is available. Chen [Che25] established a dimension-free estimate with an additional $(\log \log u)^{3/2}$ factor using perturbed reverse heat. Xiang and Zhang [XZ26] reduced the extra factor to $\log \log u$ by localizing the terminal testing discrepancy. Lu, Guo, and Fang [LGF26] removed this loss and proved the conjectured logarithmic decay. Shaposhnikov [Sha26] subsequently gave a martingale proof with the explicit coefficient $\sqrt{\kappa_a}$, where $\kappa_a := (1 + a)/(1 - a)$.

1.1 Our results

Let μ be the uniform measure on $\{-1, 1\}^n$, and let T_a be the noise operator in Eq. (19). For $0 < a < 1$, write

$$\kappa_a := \frac{1 + a}{1 - a}, \quad C_*(a) := \sup_{n \geq 1} \sup_{u > 1} \sqrt{\log u} \psi_{a,n}(u).$$

Here $\psi_{a,n}$ is defined in Eq. (20). Thus an upper bound $C_*(a) \leq B(a)$ gives $u \Pr_\mu[T_a f \geq u] \leq B(a)/\sqrt{\log u}$ for every $n \geq 1$, $u > 1$, and $f \geq 0$ with $\mathbb{E}_\mu[f] = 1$.

Define the explicit variance coefficient

$$J_a := \begin{cases} \frac{2\kappa_a(\kappa_a + 1)^2}{(3\kappa_a + 1)^2}, & 1 < \kappa_a \leq 2 + \sqrt{5}, \\ \frac{(\kappa_a + 1)^4}{4\kappa_a(\kappa_a^2 + 4\kappa_a + 1)}, & \kappa_a \geq 2 + \sqrt{5}. \end{cases} \quad (1)$$

Define $\rho_a := d(a)/a^2$, where d is the binary entropy increment in Eq. (21). Put

$$\hat{\gamma} := \frac{44989288}{100000000}, \quad c_+ := \frac{4541}{12500}, \quad c_A := \frac{357671}{1000000} < 0.3577. \quad (2)$$

The direct martingale estimate in Sections 4.5–4.6 gives a coefficient $\Gamma_B < \hat{\gamma}$, with the strict inequality certified by exact rational arithmetic. The joint entropy-cost estimate in Sections 4.8–4.10 gives c_+ . The aligned entropy clock in Sections 4.11–4.12 gives the smaller limiting constant c_A , with the finite-noise correction $B_a(\epsilon)$ defined in Eq. (76). Define

$$U_a := \min\left\{\sqrt{J_a}, \frac{\sqrt{\kappa_a \rho_a} + 1/\sqrt{2}}{2}, \hat{\gamma}\sqrt{(\kappa_a + 1/2)\rho_a} + 1, c_+\sqrt{\kappa_a + 1} + 1, 1 + \inf_{\epsilon > 0} B_a(\epsilon)\right\}. \quad (3)$$

We state our results as follows.

Theorem 1.1 (Informal version of Theorem 4.1). *For every $n \geq 1$, $0 < a < 1$, and $u > 1$, writing $\ell := \log u$, we have*

$$\psi_{a,n}(u) \leq \min\{1, U_a/\sqrt{\ell}\}.$$

For every $0 < a < 1$,

$$C_*(a) \leq U_a. \quad (4)$$

Furthermore,

$$\limsup_{a \uparrow 1} \frac{C_*(a)}{\sqrt{\kappa_a}} \leq c_A < 0.3577.$$

The previous best upper leading constant was 1 [Sha26]; we improve it to $c_A < 0.3577$.

Theorem 1.2 (Informal version of Theorem 5.1). *Let $\mathcal{L}(a)$ be the best lower bound on $C_*(a)$ obtained from point-mass inputs, optimizing over the dimension and tail threshold, as defined in Eq. (87). For every $0 < a < 1$, we have $C_*(a) \geq \mathcal{L}(a)$, and*

$$\lim_{a \uparrow 1} \frac{\mathcal{L}(a)}{\sqrt{\kappa_a}} = \sqrt{\frac{\log 2}{2e}} > 0.3570.$$

To our knowledge, no previous lower bound on this leading constant was known.

Organization. Section 2 explains the four preceding proof approaches and the ideas behind our upper and lower bounds. Section 3 fixes notation. Section 4 contains the upper-bound proof. Section 5 contains the lower-bound proof. Appendix A gives the finite coefficient construction for the joint entropy-cost comparison. Appendix B gives the finite coefficient construction for the aligned entropy-clock comparison.

(a) Upper bounds				
Year	Authors	Reference	Numerator	Dep. as $a \uparrow 1$
2025	Chen	[Che25]	$\frac{C(\log \log u)^{3/2}}{(1-a)^2}$	$O(\kappa_a^2)(\log \log u)^{3/2}$
2026	Xiang and Zhang	[XZ26]	$C_a \log \log u$	Not quantified
2026	Lu, Guo, and Fang	[LGF26]	$C\kappa_a^2 \sqrt{\frac{\log \kappa_a}{\kappa_a - 1}}$	$O(\kappa_a^{3/2} \sqrt{\log \kappa_a})$
2026	Shaposhnikov	[Sha26]	$\sqrt{\kappa_a}$	$1.0000\sqrt{\kappa_a}$
2026	This paper	Theorem 1.1	U_a	$0.3577\sqrt{\kappa_a}$
(b) Lower bounds				
Year	Authors	Reference	LB on $C_*(a)$	Dep. as $a \uparrow 1$
2026	This paper	Theorem 1.2	$\mathcal{L}(a)$	$0.3570\sqrt{\kappa_a}$

Table 1: Upper and lower bounds for Boolean convolution. Dep. denotes the asymptotic dependence as $a \uparrow 1$. Upper and lower decimal coefficients are rounded up and down, respectively, to four decimal places and give valid bounds for a sufficiently close to one; the cited theorems give the exact constants. (a) For $0 < a < 1$, $n \geq 1$, and $f \geq 0$ with $\mathbb{E}_\mu[f] = 1$, the bounds have the form $\Pr_\mu[T_a f \geq u] \leq \text{numerator}/(u\sqrt{\log u})$. Chen’s and Xiang–Zhang’s bounds hold for $u > e^3$; the others hold for $u > 1$. Chen’s and Xiang–Zhang’s heat parameter satisfies $a = e^{-\tau}$. The Chen row uses version 2 of [Che25], revised in May 2026; its year records the original preprint. The strict-tail estimates of Chen, Xiang–Zhang, and Shaposhnikov extend to non-strict tails by applying them at $v < u$ and taking $v \uparrow u$. In the Chen and Lu–Guo–Fang rows, C denotes an unspecified absolute constant that may differ between rows. In the Xiang–Zhang row, C_a depends only on a , with growth unquantified in Theorem 1.4 of [XZ26]. The extra logarithmic factors in the first two rows prevent these estimates alone from bounding $C_*(a)$; the remaining rows are uniform in u and also bound $C_*(a)$. The final two rows have explicit constants, with U_a defined in Eq. (3). (b) The point-mass profile $\mathcal{L}(a)$ in Eq. (87) satisfies $C_*(a) \geq \mathcal{L}(a)$ for every $0 < a < 1$. Its exact leading constant is optimal within the point-mass family. The final columns describe the displayed bounds; the optimal leading constant over arbitrary inputs remains undetermined.

2 Technique overview

Section 2.1 separates Chen’s proof into four steps: reduction to information bands, reverse-heat coupling, terminal comparison, and time averaging. Section 2.2 explains how Xiang–Zhang localize the third step while retaining the other ingredients. Section 2.3 describes Lu–Guo–Fang’s new coupling and band contraction. Section 2.4 presents Shaposhnikov’s alternative coordinate-martingale proof. Section 2.5 identifies our changes to its stopping, score, and final moment estimate; Section 2.6 explains the point-mass construction and its optimality within fixed-block families. Throughout, $f \geq 0$ has mean one, $F := T_a f$, $\ell := \log u$, and $P_t := T_{e^{-t}}$.

2.1 Chen: perturbed reverse heat

We organize Chen’s argument [Che25, Section 3] into four major steps.

Step 1: reduce tails to information bands. Under $d\nu_t := P_t f d\mu$, define the density-weighted profile

$$\mathfrak{A}_t(I) := \nu_t\{\log P_t f \in I\}, \quad u \Pr_\mu[F > u] \leq \sum_{j \geq 0} e^{-j} \mathfrak{A}_{-\log a}((\ell + j, \ell + j + 1]). \quad (5)$$

It suffices to control the mass of each unit information interval.

Step 2: construct the perturbed reverse heat. The reverse process V_t has marginal law ν_{T-t} and edge rate

$$q_t(x, x^{(i)}) := \frac{P_{T-t}f(x^{(i)})}{2P_{T-t}f(x)}. \quad (6)$$

Starting at time θ , couple it to W_t with perturbed jump rates, stopped at suitable information barriers.

Step 3: compare the terminal laws and levels. Choose T so that $T_\circ := T + \log a > 1$. For $u > e^3$, put $\alpha := \frac{1}{2} \log \ell + 1$, and $R_\theta := [\ell - \log P_{T-\theta}f(V_\theta)]_+$. For $\theta \in [T_\circ - 1, T_\circ]$, the global discrepancy satisfies

$$D_\theta := d_{\text{TV}}(\mathcal{L}(V_{T_\circ}), \mathcal{L}(W_{T_\circ})) \leq C_a(\alpha\sqrt{I_\theta} + \alpha^2 I_\theta), \quad I_\theta := \mathbb{E}\left[\frac{\mathbf{1}_{\{R_\theta \geq \alpha\}}}{R_\theta + 1}\right]. \quad (7)$$

Boolean-bridge and stopped score estimates give this dimension-free bound. Approximate monotonicity shifts the perturbed terminal logarithmic density down by one, apart from starting-band and exceptional-path errors. Combining these two comparisons controls the target band.

Step 4: average in time and return to the tail. The occupation estimate

$$\int_0^\infty \mathfrak{A}_t((\ell, \ell + 1]) dt \lesssim \ell^{-1} \quad (\ell > 2) \quad (8)$$

gives $\int_{T_\circ-1}^{T_\circ} I_\theta d\theta \lesssim \alpha/\ell$. Thus the square-root term in Eq. (7) costs $\alpha^{3/2}/\sqrt{\ell}$. Using Eq. (5) yields $\Pr_\mu[F > u] \leq C(1-a)^{-2}(\log \log u)^{3/2}/(u\sqrt{\log u})$ for $u > e^3$.

2.2 Xiang–Zhang: localizing the terminal discrepancy

Retained steps. Xiang–Zhang [XZ26, Sections 3–4] keep Chen’s band reduction (5), reverse process (6), perturbation, and approximate monotonicity. They also use the same occupation estimate (8). Their change is to the discrepancy part of Chen’s Step 3 and its averaging in Step 4.

Replace the global estimate by a layerwise estimate. With the same R_θ and α , set $G_r(\theta) := \{r \leq R_\theta < r + 1\}$. Carry its indicator through the Duhamel formula, terminal Boolean bridge, and Doob-transform energy estimate. In place of the global square root in Eq. (7), the cost of a layer is

$$C_a\left(\frac{\alpha}{\sqrt{r}} + \frac{\alpha^2}{r}\right) \Pr[G_r(\theta)]. \quad (9)$$

The layer probability stays outside the square root.

Average each layer before summing. Equation (8) now gives

$$\frac{\alpha}{\ell} \sum_{\alpha \leq r \leq \ell/2} r^{-1/2} + \frac{\alpha^2}{\ell} \sum_{\alpha \leq r \leq \ell/2} r^{-1} \lesssim \frac{\alpha}{\sqrt{\ell}}. \quad (10)$$

The remaining starting regions use the same stopped comparison. The unchanged band reduction (5) gives $C_a \log \log u/(u\sqrt{\log u})$. The remaining logarithm comes from $\alpha \asymp \log \ell$ in the retained perturbation and stopping buffer.

2.3 Lu–Guo–Fang: power coupling and band contraction

Retained steps. Lu–Guo–Fang [LGF26, Sections 3–4] retain Chen’s band reduction (5), reverse marginal (6), and time averaging (8). They adapt Xiang–Zhang’s localized bridge method behind Eq. (9) to a new coupling; the perturbed rates and stopped energy estimate must be rechecked.

Change the coupling and terminal displacement. This replaces Chen’s Step 2 and the displacement part of Step 3. For $Y := P_{T-t}f(x^{(i)})/P_{T-t}f(x)$, freeze at the starting time an exponent δ proportional to $1/(R_\theta + 1)$, with a fixed numerator. The common and extra flip rates are

$$\begin{cases} Y/2, & (1 - Y^\delta)/2 \text{ for the second coordinate, } Y < 1, \\ Y^{1-\delta}/2, & (Y - Y^{1-\delta})/2 \text{ for the first, } Y \geq 1. \end{cases}$$

A switched exponential weight restores the correct reverse rate for the perturbed coordinate. Its supermartingale comparison replaces approximate monotonicity by a contraction among terminal bands, with exponentially decaying weights on higher bands.

Close Step 4 without a growing buffer. The localized discrepancy and stopped score energy are averaged using Eq. (8). The band recurrence contracts with a fixed parameter, removing the growing α in Eq. (10) and yielding

$$\mathfrak{A}_{-\log a}((\ell, \ell + 1]) \lesssim \kappa_a^2 \sqrt{\frac{\log \kappa_a}{\kappa_a - 1}} \ell^{-1/2}.$$

Equation (5) then gives the conjectured decay.

2.4 Shaposhnikov: a stopped coordinate martingale

Shaposhnikov [Sha26, Section 3] gives an alternative argument. The density-weighted viewpoint of Chen’s Step 1 remains, but the band reduction, perturbed coupling, and time averaging in Eqs. (5)–(8) are replaced by the following three steps.

Step 1: reveal coordinates and stop the likelihood. Under Q , sample $Z \sim f\mu$ and then its noisy observation $X \sim F\mu$. For $F_i := \mathbb{E}_\mu[F \mid X_1, \dots, X_i]$,

$$F_i/F_{i-1} = 1 + b_i X_i, \quad b_i := \mathbb{E}_Q[X_i \mid X_1, \dots, X_{i-1}]. \quad (11)$$

Stop at the first crossing of u , or at n , and retain $v_i := b_i \mathbf{1}_{\{i \leq \sigma\}}$. On $\{F > u\}$, $M := \sum_i v_i X_i \geq \log F_\sigma \geq \ell$.

Step 2: transfer the tail to a centered score. The edge-flip identity gives

$$S := \sum_i \frac{v_i(X_i - aZ_i)}{1 - aZ_i \operatorname{sgn}(v_i)}, \quad u\mathbb{E}_\mu[\mathbf{1}_{\{F > u\}}M] \leq \mathbb{E}_Q[\mathbf{1}_{\{F > u\}}S]. \quad (12)$$

Conditionally on Z , the increments are martingale differences, and $\mathbb{E}_Q[S^2] \leq \kappa_a \mathbb{E}_Q[\sum_i v_i^2]$.

Step 3: use entropy and a second moment. The entropy inequality $d(b) \geq b^2/2$ and the overshoot bound $F_\sigma \leq u(1 + a)$ give $\mathbb{E}_Q[S^2] \leq 4\kappa_a \ell$ for $u \geq e$. Centering the indicator at $1/2$ yields

$$u\ell \Pr_\mu[F > u] \leq \frac{1}{2} \sqrt{\mathbb{E}_Q[S^2]} \leq \sqrt{\kappa_a \ell}. \quad (13)$$

Markov’s inequality covers smaller thresholds, giving $\sqrt{\kappa_a}$.

2.5 Our upper bound: exact capping and the aligned entropy clock

Retained framework. Our proof in Section 4 retains Shaposhnikov’s joint law, coordinate likelihood identity (11), and edge-flip identity behind Eq. (12). We modify his Step 1 and Step 3; the edge identity in Step 2 remains valid for a free predictable score.

Step 1: replace the crossing rule by exact capping. We keep the predictable score free. This allows the score to follow the logarithmic likelihood and its entropy cost more accurately. An exact cap attenuates the final coefficient so that the likelihood reaches its barrier without an overshoot. The conditional variance is evaluated before taking a worst-case bound, giving the scalar coefficient $H(a)$ and its explicit upper bound J_a . This already improves the leading coefficient to $1/2$.

Step 2: charge the score to a pathwise entropy budget. The sharper argument uses an entropy-drift score. For conditional bias s and attenuation $r \leq s$, its entropy charge is

$$e(s, r) := s \operatorname{arctanh} r + \frac{1}{2} \log(1 - r^2), \quad d(s) = e(s, s). \quad (14)$$

Simultaneous likelihood and entropy barriers retain charges e_i with $\sum_i e_i \leq \ell$ pathwise. Writing $y := u \Pr_\mu[F \geq u]$, the edge comparison and the discarded remainder yield

$$y\ell \leq \mathbb{E}_Q[S_+] + \sqrt{y\ell}, \quad \alpha_i := |v_i| = e_i/s_i, \quad \sum_i \alpha_i^2 \leq \rho_a \ell. \quad (15)$$

This replaces the second-moment closure in Eq. (13) by a positive-part estimate for a centered score with a deterministic entropy budget; the remainder records the cost of the modified stopping. Each two-point score increment is embedded as a martingale that rises continuously and has at most one downward jump. A scalar majorant first bounds its positive part using quadratic variation. The joint entropy comparison then retains the relation between the jump size and its entropy charge, improving the constant beyond this variance-only estimate.

Step 3: retain alignment in the entropy clock. For the final improvement, split the score according to whether the latent sign agrees with the predictable bias. The unaligned part has variance at most $\rho_a \ell / \kappa_a$ and contributes only a lower-order term. After dividing the aligned score by $K := \kappa_a + 1$, charge it the clock

$$\mathcal{A}_n := \frac{1}{K} \sum_i \frac{2e_i I_i}{1 + s_i}, \quad I_i := \mathbf{1}_{\{Z_i \operatorname{sgn}(b_i) = 1\}}. \quad (16)$$

The entropy charges from Eq. (14) are retained. Two exponential martingale estimates control the clock (16) around $(1 + 1/\kappa_a)\ell/K$. More precisely, for a centered W ,

$$K\mathcal{A}_n \leq (1 + 1/\kappa_a)\ell + W, \quad \mathbb{E}_Q[e^{tW}] \leq e^{\gamma_a \ell t^2/2}, \quad \gamma_a := \rho_a(1 + 1/\kappa_a)^2. \quad (17)$$

Stopping at a slightly larger deterministic budget therefore has a quantitatively bounded residual cost. The clock continues to consume its charge after a stage jump. This preserves the entropy accounting when the score has already stopped moving and is used explicitly in the potential’s drift calculation.

Step 4: replace Cauchy–Schwarz by a potential comparison. A potential $U(t, x)$ majorizing x_+ bounds the stopped score when its jump-compensator drift is nonpositive. On each jump bin the sufficient test is

$$p(h)\{U(t, x - h) - U(t, x) + hU_x(t, x)\} \leq h^2 U_t(t, x), \quad (18)$$

where the certified cost $p(h)$ converts the jump size into a clock charge. For the fine potential, affine chords majorize the convex inverse-entropy cost. The construction checks this inequality

jointly in spatial state and jump size, including arbitrarily small jumps and every time within each interval. Rational spline potentials cover bounded entropy times; a Gaussian comparison covers the unbounded range. Appendices A and B contain the finite constructions. Applying Eq. (18), returning from the stopped clock via Eq. (17), and adding the unaligned part bound the positive part in Eq. (15). Then optimizing the clock allowance gives $C_*(a) \leq 1 + \inf_{\epsilon > 0} B_a(\epsilon)$. Together with the earlier branches this is U_a , with limiting coefficient $c_A = 0.357671$. The estimates hold before the suprema over dimension, density, and threshold are taken.

2.6 Our lower bound: point masses and fixed blocks

Section 5 starts from the normalized point mass $f_n := 2^n \mathbf{1}_{\{(1, \dots, 1)\}}$. Its noisy image is the product $T_a f_n(x) = \prod_i (1 + ax_i)$, so every output level and its tail mass are explicit. At the top level $u := (1 + a)^n$, only one vertex belongs to the tail. With $p := (1 + a)/2$, its contribution is

$$u \sqrt{\log u} \Pr_{\mu}[T_a f_n \geq u] = \sqrt{n \log(1 + a)} p^n.$$

Optimizing the real dimension gives $n_* := -1/(2 \log p)$; the best integer is one of its two neighbors. As $a \uparrow 1$, $n_* \sim \kappa_a/2$, and this construction gives the coefficient $\sqrt{\log 2/(2e)}$. All other attained levels correspond to Hamming balls and give the explicit profile $\mathcal{L}(a)$.

To show that this coefficient is exact within the family, the proof controls all dimensions and thresholds through a uniform binomial-atom estimate. If $b_m(\theta)$ is the largest atom of $\text{Bin}(m, \theta)$, then

$$\limsup_{\theta \downarrow 0} \sup_{m \geq 1} \sqrt{m\theta} b_m(\theta) = \frac{1}{\sqrt{2e}}.$$

The proof treats vanishing, finite, and diverging $m\theta$ separately; the middle regime reduces to Poisson atoms. An antichain argument transfers this bound to sums of block log-likelihoods. For a fixed normalized indicator block, the resulting coefficient is controlled by its entropy h_B and outward edge boundary β_B :

$$\sqrt{\frac{h_B}{2e\beta_B}} \leq \sqrt{\frac{\log 2}{2e}}, \quad h_B \leq \beta_B \log 2.$$

Here the second inequality is the cube edge-isoperimetric bound. Nonflat fixed blocks contribute zero in this normalization. This proves optimality within fixed-block tensor families while leaving arbitrary densities outside that conclusion.

3 Preliminaries

All logarithms are natural. For $n \geq 1$, write $[n] := \{1, \dots, n\}$, $G_n := \{-1, 1\}^n$, and $\mu := \mu_n$ for the uniform probability measure on G_n . We use $\mathbf{1}_A$ for the indicator of an event A , $x_+ := \max\{x, 0\}$, and $\text{sgn}(0) := 0$. The vector $x^{(i)}$ is obtained from x by flipping coordinate i . Expectations and variances carry a subscript when their probability law needs to be specified.

3.1 Noise and weak tail profiles

Let μ denote the uniform probability measure on $G_n := \{-1, 1\}^n$. For $0 < a < 1$, define the noise operator

$$T_a f(x) := 2^{-n} \sum_{z \in G_n} f(z) \prod_{i=1}^n (1 + ax_i z_i). \quad (19)$$

For $u > 1$, put

$$\psi_{a,n}(u) := \sup_{\mu} \{u \Pr[T_a f \geq u] : f \geq 0, \mathbb{E}_{\mu}[f] = 1\}. \quad (20)$$

The operator preserves nonnegativity and expectation. Its product kernel is strictly positive for $0 < a < 1$, so $T_a f > 0$ whenever $f \geq 0$ is not identically zero. We normally normalize $\mathbb{E}_{\mu}[f] = 1$; an estimate for a general nonzero input follows by applying it to $f/\mathbb{E}_{\mu}[f]$. Our heat-semigroup convention is $P_t := T_{e^{-t}}$. We write

$$\kappa_a := \frac{1+a}{1-a}, \quad C_*(a) := \sup_{n \geq 1} \sup_{u > 1} \sqrt{\log u} \psi_{a,n}(u).$$

For tensor products, the noise operator acts independently on the factors; $g^{\otimes m}$ denotes the product of m copies of a block density g . The notation $\text{Bin}(m, p)$ denotes a binomial random variable.

3.2 Entropy and martingale notation

Define the binary entropy increment

$$d(s) := \frac{1+s}{2} \log(1+s) + \frac{1-s}{2} \log(1-s), \quad |s| < 1. \quad (21)$$

It satisfies $d(0) = d'(0) = 0$ and $d''(s) = 1/(1-s^2)$, hence $d(s) \geq s^2/2$. We extend d continuously to $[-1, 1]$, with $d(1) = d(-1) = \log 2$, and put $\rho_a := d(a)/a^2$. For a density h relative to a probability law ν , its entropy is $\text{Ent}_{\nu}(h) := \mathbb{E}_{\nu}[h \log h]$, with $0 \log 0 := 0$. More generally we subtract $\mathbb{E}_{\nu}[h] \log(\mathbb{E}_{\nu}[h])$ when h is not normalized. A coefficient at step i is predictable if it is measurable with respect to the information available before that step. For a martingale L , $\langle L \rangle$ denotes its predictable quadratic variation; in discrete time its increments are the conditional second moments of the martingale differences. We use $\varphi(x) := (2\pi)^{-1/2} e^{-x^2/2}$ for the standard Gaussian density.

3.3 The joint law and coordinate revelation

Fix $n \geq 1$, $0 < a < 1$, $u > 1$, and an arbitrary nonnegative density f with $\mathbb{E}_{\mu}[f] = 1$. Write $F := T_a f$, $\ell := \log u$, and $A := \{F \geq u\}$. Under Q , give Z distribution $f\mu$ and, conditionally on Z , sample independent coordinates satisfying

$$\Pr_Q[X_i = x_i \mid Z] := \frac{1 + aZ_i x_i}{2}.$$

Then X has distribution $F\mu$. The kernel is strictly positive, so $F > 0$ and every observed prefix has positive probability. Let $\mathcal{F}_i := \sigma(X_1, \dots, X_i)$ and

$$b_i := \mathbb{E}_Q[X_i \mid \mathcal{F}_{i-1}] = a \mathbb{E}_Q[Z_i \mid \mathcal{F}_{i-1}], \quad |b_i| \leq a.$$

For $F_i := \mathbb{E}_{\mu}[F \mid \mathcal{F}_i]$, Bayes' rule gives $F_i/F_{i-1} = 1 + b_i X_i$ and $F_0 = 1$.

We write $y := u \Pr_{\mu}[A]$ for the scaled tail mass, so $0 \leq y \leq 1$ by Markov's inequality. The observed-prefix filtration $\mathcal{F}_i$ is distinct from $\sigma(Z, X_1, \dots, X_i)$, in which the latent vector is revealed initially. Conditional independence of the X_i holds given Z , even when the coordinates of Z are dependent. Both filtrations will be used explicitly in the upper-bound proof.

4 Upper bound

Section 4.1 states and proves the main theorem using the auxiliary results developed below. Section 4.2 defines the scalar variance coefficient and bounds it by the explicit coefficient J_a . Section 4.3 develops the predictable edge comparison and its variance and entropy identities. Section 4.4 constructs the exact likelihood cap and entropy stopping rule and derives the resulting score bounds. Section 4.5 embeds the score in a downward-jump martingale and establishes the verification principle and scalar curve for the comparison. Section 4.6 constructs the positive-part comparison profile and certifies its constant by exact arithmetic. Section 4.7 deduces the corresponding convolution bound and proves the Gaussian comparison used at large entropy times. Section 4.8 normalizes the entropy score and establishes a verification principle that retains the joint constraints on jump size and entropy cost. Section 4.9 constructs spline potentials and gives finite polynomial tests for their drift inequalities. Section 4.10 verifies the objective bounds uniformly over entropy time and completes the joint entropy-cost estimate. Section 4.11 separates the aligned and unaligned scores and proves concentration estimates for the aligned entropy clock. Section 4.12 applies the aligned-clock potential comparison and controls the stopping remainder to prove the aligned bound.

4.1 Main theorem and proof

We first state the full tail estimate and the resulting bound on the optimal uniform constant. The proof assembles the comparisons established in the later subsections.

Theorem 4.1 (Formal version of Theorem 1.1). *For $0 < a < 1$, set $\kappa_a := (1 + a)/(1 - a)$, let J_a , U_a , and c_A be defined in Eq. (1), Eq. (3), and Eq. (2), and put*

$$C_*(a) := \sup_{n \geq 1} \sup_{u > 1} \sqrt{\log u} \psi_{a,n}(u).$$

Let H , $\mathcal{R}$, and Θ_a be defined in Eq. (24), Eq. (25), and Eq. (75), respectively. For every integer $n \geq 1$, every $0 < a < 1$, every $u > 1$, and every function $f : G_n \rightarrow [0, \infty)$ with $\mathbb{E}_\mu[f] > 0$, writing $\ell := \log u$, we have

$$\Pr_\mu[T_a f \geq u \mathbb{E}_\mu[f]] \leq \frac{1}{u} \min\{1, \mathcal{R}(H(a)/\ell), U_a/\sqrt{\ell}, \Theta_a(\ell)\}. \quad (22)$$

For every $0 < a < 1$,

$$C_*(a) \leq U_a. \quad (23)$$

Moreover,

$$\limsup_{a \uparrow 1} \frac{C_*(a)}{\sqrt{\kappa_a}} \leq c_A < 0.3577.$$

Proof. Apply Corollary 4.23 to $f/\mathbb{E}_\mu[f]$, with $c_j = c_+$. This proves Eq. (22) and Eq. (23).

The sharper threshold-dependent estimate is also retained:

$$\psi_{a,n}(u) \leq \min\left\{\mathcal{R}(H(a)/\ell), \frac{\sqrt{\kappa_a \rho_a} + 1/\sqrt{2}}{2\sqrt{\ell}}, \frac{\Gamma_B \sqrt{(\kappa_a + 1/2)\rho_a} + 1}{\sqrt{\ell}}, \frac{c_+ \sqrt{\kappa_a + 1} + 1}{\sqrt{\ell}}, \Theta_a(\ell)\right\},$$

where $\ell := \log u$ and $\mathcal{R}$ is defined in Eq. (25).

The aligned branch satisfies $B_a(\epsilon) \geq c_A \sqrt{(\kappa_a + 1)(1 + 1/\kappa_a)}$ for every $\epsilon > 0$. The explicit choice in Eq. (86) gives the matching normalized upper limit c_A . The four earlier branches have normalized limits $1/2$, $\sqrt{\log 2}/2$, $\hat{\gamma}\sqrt{\log 2}$, and c_+ , all strictly larger. Thus $U_a/\sqrt{\kappa_a} \rightarrow c_A$. These limits concern the explicit upper bound; they do not assert existence of a normalized limit for $C_*(a)$. All estimates hold for every admissible density, dimension, and threshold. $\square$

4.2 Scalar coefficients and variance bounds

Use the binary entropy increment d from Eq. (21). Define

$$H(a) := \frac{1}{4(1-a^2)} \max_{0 \leq s \leq a} \frac{(1+a^2+2s) \log^2(1+s)}{d(s)}. \quad (24)$$

The quotient at zero is defined by continuity to be $2(1+a^2)$, since $d(s)/s^2 \rightarrow 1/2$ and $\log(1+s)/s \rightarrow 1$. Thus the maximum exists. For $t \geq 0$, put

$$\mathcal{R}(t) := \begin{cases} \sqrt{t}, & 0 \leq t \leq 1/4, \\ 4t/(1+4t), & t \geq 1/4. \end{cases} \quad (25)$$

Proposition 4.6 proves $\psi_{a,n}(u) \leq \mathcal{R}(H(a)/\log u)$ and $C_*(a) \leq \sqrt{H(a)}$. The next lemma bounds $H(a)$ by the explicit coefficient J_a .

Lemma 4.2. *For $0 < a < 1$, the coefficient J_a in Eq. (1) satisfies*

$$H(a) \leq J_a < \frac{\kappa_a + 1}{4}. \quad (26)$$

Moreover, $J_a = \kappa_a/4 + O(1/\kappa_a)$ as $a \uparrow 1$.

Proof. We first prove the scalar inequality

$$\frac{\log^2(1+s)}{d(s)} \leq \frac{8}{(2+s)^2} \quad (0 < s < 1). \quad (27)$$

Let $g(s) := s + s^3/12 - (1+s/2) \log(1+s)$. Then $g(0) = g'(0) = 0$ and $g''(s) = s^2(2+s)/(2(1+s)^2) \geq 0$, so $g(s) \geq 0$. The positive power series $d(s) = \sum_{j \geq 1} s^{2j}/((2j)(2j-1))$ gives

$$2d(s) \geq s^2 + s^4/6 + s^6/15 \geq (s + s^3/12)^2 \geq (1 + s/2)^2 \log^2(1+s).$$

The first step retains the first three series terms; the second uses $1/15 \geq 1/144$; the third uses $g(s) \geq 0$ and nonnegativity. This proves Eq. (27), with equality in the continuous extension at zero.

Write $D := 1 - a^2$. Substituting Eq. (27) into Eq. (24) yields

$$H(a) \leq \frac{2}{D} \max_{0 \leq s \leq a} \frac{1 + a^2 + 2s}{(2+s)^2}.$$

The derivative of the quotient is $2(D-s)/(2+s)^3$. Thus its maximum occurs at $s_* := \min\{a, D\}$ and the last bound equals

$$J_a = \frac{2(1 + a^2 + 2s_*)}{(1 - a^2)(2 + s_*)^2}.$$

The two cases $a \leq D$ and $a \geq D$ give Eq. (1) after substituting $a = (\kappa_a - 1)/(\kappa_a + 1)$.

In the first case, the ratio of J_a to $1/(2(1-a))$ is $4(1+a)/(2+a)^2 < 1$. In the second case, the ratio is $4/((1+a)(3-a^2)) < 1$: indeed, $(1+a)(3-a^2) - 4 = (1-a)(a^2 + 2a - 1) > 0$ for $(\sqrt{5}-1)/2 \leq a < 1$. This proves the strict inequality in Eq. (26). Finally, in the second branch,

$$J_a - \frac{\kappa_a}{4} = \frac{5\kappa_a^2 + 4\kappa_a + 1}{4\kappa_a(\kappa_a^2 + 4\kappa_a + 1)} = O(1/\kappa_a),$$

which proves the asserted expansion. $\square$

The preceding estimates give the preliminary upper leading constant $1/2$. The subsequent comparisons, assembled in Section 4.12, prove the stronger upper bounds in Theorem 4.1.

4.3 Predictable scores and entropy identities

Use Q , X , Z , F , A , ℓ , $\mathcal{F}_i$, b_i , and F_i from Section 3.

Lemma 4.3 (Predictable edge comparison). *Let v_i be arbitrary real $\mathcal{F}_{i-1}$ -measurable coefficients. Put*

$$M := \sum_{i=1}^n v_i X_i, \quad S_i := \frac{v_i(X_i - aZ_i)}{1 - aZ_i \operatorname{sgn}(v_i)}, \quad S := \sum_{i=1}^n S_i,$$

with $S_i := 0$ when $v_i = 0$. Then $\mathbb{E}_Q[S] = 0$ and

$$u\mathbb{E}_\mu[\mathbf{1}_A M] \leq \mathbb{E}_Q[\mathbf{1}_A S].$$

Also

$$\mathbb{E}_Q[S_i^2 \mid Z, \mathcal{F}_{i-1}] = v_i^2 \frac{1 + aZ_i \operatorname{sgn}(v_i)}{1 - aZ_i \operatorname{sgn}(v_i)}, \quad \mathbb{E}_Q[S^2] = \sum_i \mathbb{E}_Q[S_i^2]. \quad (28)$$

If v_i has the sign of b_i and $s_i := |b_i|$, then

$$\mathbb{E}_Q[S_i^2 \mid \mathcal{F}_{i-1}] = v_i^2 \frac{1 + a^2 + 2s_i}{1 - a^2}. \quad (29)$$

Proof. Conditionally on Z and the past, X_i has mean aZ_i . This proves the conditional mean-zero assertion and the first variance identity. Orthogonality of the increments follows by conditioning on the filtration revealing both Z and the observed prefix. The identity

$$\frac{1 + a\epsilon}{1 - a\epsilon} = \frac{1 + a^2 + 2a\epsilon}{1 - a^2}, \quad \epsilon \in \{-1, 1\},$$

and, on $\{v_i \neq 0\}$, the identity $a\mathbb{E}_Q[Z_i \operatorname{sgn}(v_i) \mid \mathcal{F}_{i-1}] = s_i$ prove Eq. (29). When $v_i = 0$, both sides of that variance formula are zero.

For completeness, write $x^{(i)}$ for x with coordinate i flipped. Pairing the two values of that coordinate, conditionally on Z , gives

$$\mathbb{E}_Q[S_i g(X)] = \mathbb{E}_Q[(v_i X_i)_+ (g(X) - g(X^{(i)}))].$$

Indeed,

$$S_i = (v_i X_i)_+ - \frac{1 - aZ_i X_i}{1 + aZ_i X_i} (-v_i X_i)_+,$$

and the displayed ratio is the conditional probability ratio of the flipped and unflipped vertices. The coefficient v_i is unchanged by this flip. The corresponding uniform-measure identity is

$$\mathbb{E}_\mu[v_i X_i g(X)] = \mathbb{E}_\mu[(v_i X_i)_+(g(X) - g(X^{(i)}))].$$

Apply these identities to $g := \mathbf{1}_A$, sum, and use the density F to get

$$\begin{aligned} \mathbb{E}_Q[\mathbf{1}_A S] - u \mathbb{E}_\mu[\mathbf{1}_A M] &= \mathbb{E}_\mu\left[\sum_i (v_i X_i)_+(F(X) - u)(\mathbf{1}_A(X) - \mathbf{1}_A(X^{(i)}))\right] \\ &\geq 0. \end{aligned}$$

Each summand is nonnegative, including at $F(X) = u$. $\square$

We next record the two-parameter entropy identities used to choose and cap the score. For $0 \leq r \leq s < 1$, define

$$w(r) := -\frac{1}{2} \log(1 - r^2), \quad e(s, r) := s \operatorname{arctanh}(r) - w(r).$$

The value at $r = 0$ is zero. Notice that $d(s) = e(s, s)$.

Lemma 4.4 (Scalar bounds). *For $0 \leq r \leq s \leq a < 1$,*

$$d(r) \leq e(s, r) \leq d(s), \quad d(s) \geq \frac{s^2}{2}, \quad \frac{d(s)}{s^2} \leq \frac{d(a)}{a^2} = \rho_a \quad (30)$$

when $s > 0$, with the continuous quotient $d(s)/s^2 = 1/2$ at zero. For $s > 0$, put $v := e(s, r)/s$. Then

$$v^2 \leq \rho_a e(s, r), \quad (1 - s^2)v^2 \leq \frac{1}{2} e(s, r), \quad \frac{(1 - s^2)w(r)^2}{s^2} \leq \frac{1}{2} e(s, r). \quad (31)$$

Moreover $0 \leq v \leq d(a)/a$.

Proof. The derivative of $e(s, r)$ with respect to r is $(s - r)/(1 - r^2) \geq 0$ for $0 \leq r \leq s$, and the derivative with respect to s is $\operatorname{arctanh}(r) \geq 0$. These give the first bounds in Eq. (30). Integrating $d''(s) = 1/(1 - s^2)$ twice gives the positive series

$$d(s) = \sum_{j \geq 1} \frac{s^{2j}}{(2j)(2j - 1)}.$$

It proves the other assertions in Eq. (30), monotonicity of $d(s)/s$, and

$$d(s) \leq \frac{s^2}{2(1 - s^2)}.$$

The first two estimates in Eq. (31) follow by using $e(s, r) \leq d(s)$ in $v^2 = e(s, r)^2/s^2$.

To prove the last estimate, the inequality $2 \log t \leq t - t^{-1}$ for $t \geq 1$ follows by differentiating its difference. Substitute $t := (1 - r^2)^{-1/2}$ and square to obtain

$$(1 - r^2)w(r)^2 \leq \frac{r^4}{4}.$$

Since $(1 - s^2)/s^2$ decreases with s ,

$$\frac{(1 - s^2)w(r)^2}{s^2} \leq \frac{(1 - r^2)w(r)^2}{r^2} \leq \frac{r^2}{4} \leq \frac{d(r)}{2} \leq \frac{e(s, r)}{2}.$$

When $r = 0$ both sides of the required estimate are zero. $\square$

4.4 Exact likelihood capping and entropy stopping

The first construction has only a likelihood barrier. Set $G_0 := 1$. If $G_{i-1} = u$, set $r_i := 0$. Otherwise, writing $s_i := |b_i|$, put

$$r_i := \min\{s_i, \frac{u}{G_{i-1}} - 1\}, \quad t_i := \text{sgn}(b_i)r_i, \quad G_i := G_{i-1}(1 + t_i X_i). \quad (32)$$

When $b_i = 0$ use $t_i := 0$. All these choices are predictable.

Lemma 4.5 (Capped likelihood). *The construction satisfies $0 < G_i \leq u$. Until the barrier is reached, $G_i \geq F_i$; once it is reached $G_i = u$ forever. In particular, $G_n = u$ on A . Finally,*

$$\mathbb{E}_Q[\sum_i e(s_i, r_i)] = \mathbb{E}_Q[\log G_n] \leq \ell. \quad (33)$$

Proof. Suppose $G_{i-1} < u$ and $G_{i-1} \geq F_{i-1}$. On an outcome opposite to $\text{sgn}(b_i)$,

$$G_i = G_{i-1}(1 - r_i) \geq F_{i-1}(1 - s_i) = F_i.$$

On an aligned outcome either $r_i = s_i$, giving $G_i \geq F_i$, or the likelihood cap is active and $G_i = u$. This proves the domination assertion by induction and also proves that a terminal path in A reaches the barrier. The upper and lower bounds for G_i follow directly from $r_i \leq a < 1$ and the cap. Conditionally on the past, $Y_i := \text{sgn}(b_i)X_i$ has mean s_i when $s_i > 0$, so

$$\mathbb{E}_Q[\log(1 + r_i Y_i) \mid \mathcal{F}_{i-1}] = e(s_i, r_i).$$

Summing gives Eq. (33). Zero coefficients cause zero increments on both sides. $\square$

Use the coefficient $H(a)$ defined in Eq. (24).

Proposition 4.6 (Overshoot-free upper bound). *For all admissible a, n, u, f , with $y := u \Pr_\mu[A]$ and $q := \Pr_Q[A]$,*

$$y\sqrt{\ell} \leq 2\sqrt{H(a)q(1-q)}. \quad (34)$$

Consequently

$$\psi_{a,n}(u) \leq \mathcal{R}\left(\frac{H(a)}{\ell}\right), \quad C_*(a) \leq \sqrt{H(a)}. \quad (35)$$

Proof. Use Eq. (32) and the logarithmic coefficients $v_i := \text{sgn}(b_i) \log(1 + r_i)$. For either sign of Y_i , $\log(1 + r_i)Y_i \geq \log(1 + r_i Y_i)$. Hence $M \geq \log G_n = \ell$ on A .

The exact conditional variance is

$$\mathbb{E}_Q[S_i^2 \mid \mathcal{F}_{i-1}] = \frac{(1 + a^2 + 2s_i) \log^2(1 + r_i)}{1 - a^2}.$$

It is essential here that s_i , not r_i , is the true conditional bias. For fixed $r > 0$, the function

$$s \mapsto \frac{1 + a^2 + 2s}{e(s, r)}$$

is decreasing on $[r, a]$: its derivative has numerator $-(1 + a^2) \text{arctanh}(r) - 2w(r) < 0$. Therefore the definition of $H(a)$ gives

$$\mathbb{E}_Q[S_i^2 \mid \mathcal{F}_{i-1}] \leq 4H(a)e(s_i, r_i).$$

This also holds when $r_i = 0$. Orthogonality and Eq. (33) imply $\mathbb{E}_Q[S^2] \leq 4H(a)\ell$. The edge comparison and centering at q now give

$$y\ell \leq \mathbb{E}_Q[(\mathbf{1}_A - q)S] \leq 2\sqrt{H(a)\ell q(1-q)}.$$

This proves Eq. (34).

For clarity, the scalar tail optimization is included. Since $F \geq u$ on A , $0 \leq y \leq q \leq 1$. Put $t := H(a)/\ell$. If $t \leq 1/4$, then $y \leq \sqrt{t}$. If $t \geq 1/4$ and $y \leq 1/2$, then $y \leq 4t/(1+4t)$. Otherwise $q \geq y > 1/2$, so $q(1-q) \leq y(1-y)$; squaring gives $y^2 \leq 4ty(1-y)$ and thus $y \leq 4t/(1+4t)$. The two branches agree at $t = 1/4$. Finally, $q(1-q) \leq 1/4$ in Eq. (34) proves the uniform bound. $\square$

We now choose the predictable score from the entropy drift. Use the same single capped process, but now set

$$v_i := \begin{cases} \operatorname{sgn}(b_i) \frac{e(s_i, r_i)}{s_i}, & s_i > 0, \\ 0, & s_i = 0. \end{cases} \quad (36)$$

Let M, S be the associated scores and put $N := M - \log G_n$.

Lemma 4.7 (A centered remainder). *The variable N is a sum of martingale differences for the observed-prefix filtration under Q . In addition,*

$$\mathbb{E}_Q[N] = 0, \quad \mathbb{E}_Q[N^2] \leq \frac{\ell}{2}, \quad \mathbb{E}_Q[S^2] \leq \kappa_a \rho_a \ell. \quad (37)$$

Proof. For $s_i > 0$, write $Y_i := \operatorname{sgn}(b_i)X_i$. Since $\log(1 + rY) = \operatorname{arctanh}(r)Y - w(r)$, the increment of N is

$$-\frac{w(r_i)}{s_i}(Y_i - s_i). \quad (38)$$

It has conditional mean zero and conditional variance at most $e(s_i, r_i)/2$ by Lemma 4.4. This proves the first two claims. Also Eq. (28) and $v_i^2 \leq \rho_a e(s_i, r_i)$ give

$$\mathbb{E}_Q[S_i^2 \mid Z, \mathcal{F}_{i-1}] \leq \kappa_a \rho_a e(s_i, r_i).$$

Sum, use orthogonality, and apply Eq. (33). $\square$

Proposition 4.8 (Entropy-drift upper bound). *For all admissible parameters,*

$$y\sqrt{\ell} \leq \sqrt{\kappa_a \rho_a q(1-q)} + \sqrt{\frac{y(1-y)}{2}}, \quad (39)$$

and therefore

$$C_*(a) \leq \frac{\sqrt{\kappa_a \rho_a} + 1/\sqrt{2}}{2}. \quad (40)$$

In particular, its upper leading constant is at most $\sqrt{\log 2}/2$.

Proof. Let $W := (u/F)\mathbf{1}_A$. Then $0 \leq W \leq 1$, $\mathbb{E}_Q[W] = y$, and $\operatorname{Var}_Q(W) \leq y(1-y)$. On A , $M = \ell + N$. The edge comparison gives

$$y\ell \leq \mathbb{E}_Q[\mathbf{1}_A S] - \mathbb{E}_Q[WN].$$

Center the first term at q and the second at y . Cauchy–Schwarz and Eq. (37) prove Eq. (39). Both scalar variances $q(1-q)$ and $y(1-y)$ are at most $1/4$, giving Eq. (40). Finally $\rho_a \rightarrow \log 2$ and $\kappa_a \rightarrow \infty$ as $a \uparrow 1$. $\square$

The next step imposes simultaneous barriers on the likelihood and the accumulated entropy. To obtain a deterministic variance budget, cap both likelihood and accumulated conditional entropy. Start with $G_0 := 1$ and $D_0 := 0$. At step i , set $s_i := \lfloor b_i \rfloor$. If $G_{i-1} = u$, $D_{i-1} = \ell$, or $s_i = 0$, use $r_i := 0$. Otherwise put

$$r_i^{\max} := \min\left\{s_i, \frac{u}{G_{i-1}} - 1\right\}$$

and take the largest $r_i \in [0, r_i^{\max}]$ such that

$$e(s_i, r_i) \leq \ell - D_{i-1}.$$

This choice exists and is unique when the entropy cap is active, by continuity and strict increase of $e(s_i, r)$ on $[0, s_i]$. Set

$$G_i := G_{i-1}(1 + \text{sgn}(b_i)r_iX_i), \quad D_i := D_{i-1} + e(s_i, r_i).$$

Use the entropy-drift score Eq. (36). Put

$$N := M - \log G_n, \quad K := M - D_n.$$

Lemma 4.9 (Double-barrier estimates). *The construction has $0 < G_i \leq u$ and $0 \leq D_i \leq \ell$ on every path. On A , at least one of $G_n = u$ and $D_n = \ell$ holds. Moreover*

$$\mathbb{E}_Q[N^2] \leq \ell/2, \quad \mathbb{E}_Q[K^2] \leq \ell/2, \quad \sum_i v_i^2 \leq \rho_a \ell \quad \text{pathwise.}$$

Consequently

$$y\ell \leq \mathbb{E}_Q[S_+] + \sqrt{y\ell}.$$

Proof. The bounds for G_i, D_i follow from their definitions. Before either barrier is reached, the induction in Lemma 4.5 still applies unless r_i is reduced by the entropy cap. In that case $D_i = \ell$ and a barrier has already been reached. Thus a path in A cannot finish with both barriers unreached.

The increments of N are still Eq. (38). Those of K are

$$\frac{e(s_i, r_i)}{s_i}(Y_i - s_i)$$

when $s_i > 0$, and zero otherwise. Lemma 4.4 bounds each conditional variance by $e(s_i, r_i)/2$. Orthogonality and $D_n \leq \ell$ give the two error bounds. The pathwise variance-coefficient estimate follows from $v_i^2 \leq \rho_a e(s_i, r_i)$.

On A , either $M = \ell + N$ or $M = \ell + K$, so

$$M \geq \ell - \sqrt{N^2 + K^2}.$$

Using again $W := (u/F)\mathbf{1}_A$, the edge comparison implies

$$y\ell \leq \mathbb{E}_Q[\mathbf{1}_A S] + \mathbb{E}_Q[W \sqrt{N^2 + K^2}].$$

The first term is at most $\mathbb{E}_Q[S_+]$. Since $\mathbb{E}_Q[W^2] \leq y$ and $\mathbb{E}_Q[N^2 + K^2] \leq \ell$, the second is at most $\sqrt{y\ell}$. $\square$

4.5 Martingale embedding and the positive-part comparison

We begin with a verification principle that turns a differential inequality for a majorant into an expectation bound for a martingale with downward jumps.

Lemma 4.10 (Martingale verification). *Let L be a bounded martingale starting at zero, obtained from finitely many compensated counting processes on a deterministic finite horizon T . Suppose its jumps are nonpositive and its predictable quadratic variation is continuous and at most a deterministic $V > 0$. Let g be continuous and let $W \in C^{1,2}((0, \infty) \times \mathbb{R})$ satisfy $W(t, x) \geq g(x)$ and*

$$W(t, x - h) - W(t, x) + hW_x(t, x) \leq h^2W_t(t, x) \quad (t > 0, x \in \mathbb{R}, h \geq 0).$$

Then

$$\mathbb{E}[g(L_T)] \leq W(V, 0). \quad (41)$$

Proof. Fix $\epsilon > 0$ and set $t_s := V + \epsilon - \langle L \rangle_s \geq \epsilon$. Let $\nu(ds, dh)$ be the predictable compensator of the measure of total downward jumps of size $h > 0$, grouping simultaneous jumps together. The compensating drift is $\int h\nu(ds, dh)$ and $d\langle L \rangle_s = \int h^2\nu(ds, dh)$. Thus the finite-variation term in $W(t_s, L_s)$ is

$$\int [W(t_s, L_{s-} - h) - W(t_s, L_{s-}) + hW_x(t_s, L_{s-}) - h^2W_t(t_s, L_{s-})]\nu(ds, dh) \leq 0.$$

This formulation also covers continuous compensators without a density in time. Boundedness of L and $t_s \geq \epsilon$ bound all derivatives on the relevant compact set; the Taylor remainders are bounded by a constant times h^2 , whose integrated compensator is at most V . Localization, followed by these bounds, therefore gives

$$\mathbb{E}[g(L_T)] \leq \mathbb{E}[W(t_T, L_T)] \leq W(V + \epsilon, 0).$$

Continuity as $\epsilon \downarrow 0$ proves Eq. (41). $\square$

To apply this principle to the edge score, we realize that score as a continuous-time martingale with downward jumps. The construction preserves its distribution and gives a pathwise bound on its predictable quadratic variation.

Lemma 4.11 (Downward-jump embedding of the edge score). *For any predictable coefficients v_i , the terminal edge score S in Lemma 4.3 can be realized as the terminal value of a bounded martingale with only downward jumps and continuous predictable quadratic variation bounded pathwise by*

$$(\kappa_a + 1/2) \sum_i v_i^2.$$

The realization preserves the joint distribution of (Z, X) and of the terminal score.

Proof. Reveal Z initially. At stage i , condition on Z and the observed prefix. If $v_i \neq 0$, put

$$\alpha := |v_i|, \quad \beta := |v_i| \frac{1 + aZ_i \operatorname{sgn}(v_i)}{1 - aZ_i \operatorname{sgn}(v_i)}.$$

The required increment is α with probability $\beta/(\alpha + \beta)$ and $-\beta$ otherwise.

Run this stage on $[i - 1, i]$, writing $t \in [0, 1]$ for its elapsed time. Starting from zero, the stage contribution is αt until a single jump to $-\beta$, after which it stays fixed. Before the jump its intensity is $\alpha/(\beta + \alpha t)$. The probability of no jump by time t is

$$\exp\left(-\int_0^t \frac{\alpha}{\beta + \alpha s} ds\right) = \frac{\beta}{\beta + \alpha t},$$

so the terminal law is the required two-point law. Its continuous drift α cancels its mean jump rate $-\alpha$. If τ is the jump time, the stage bracket is continuous and bounded by

$$\int_0^{1 \wedge \tau} \alpha(\beta + \alpha t) dt \leq \alpha\beta + \alpha^2/2 \leq (\kappa_a + 1/2)v_i^2.$$

Use independent randomization at each stage and reveal X_i from its terminal outcome. This gives the correct conditional law of X_i , hence preserves the joint law of (Z, X) even when later coefficients are adaptive. When $v_i = 0$, keep the score fixed and sample X_i from its original conditional law at the stage endpoint.

Concatenating the stages produces terminal score S on the deterministic horizon $T := n$, with the stated pathwise bracket bound. The jumps are downward and occur on disjoint stages; the intensities are at most κ_a because $\beta \geq \alpha/\kappa_a$. There are at most n jumps, and for the fixed finite problem the coefficients, scores, and brackets are bounded. $\square$

We next apply a positive-part comparison to the embedded martingale.

Let $\tau := (\sqrt{5} - 1)/2$. The next differential equation determines the space-time majorant used below. It is a definition of an exact constant, not a numerically conjectured differential inequality.

Lemma 4.12 (Existence and identities for the scalar curve). *There is a unique positive function ξ on $(0, \infty)$, continuous at zero with $\xi(0) = 0$, satisfying*

$$\xi'(r) = \frac{r}{\xi(r)} - 1 + \frac{r^2 - \xi(r)^2}{2} \quad (r > 0). \quad (42)$$

It is smooth on $(0, \infty)$ and satisfies

$$\tau r \leq \xi(r) < r, \quad \xi(r) \leq \tau r + \frac{r^3}{8}, \quad 0 < \xi'(r) < 2.$$

In particular, $\xi(r)/r \rightarrow \tau$ as $r \downarrow 0$, and ξ is a bijection of $(0, \infty)$ onto itself. Define

$$D(r) := 2 + \xi(r)^2 - r^2, \quad I(r) := \int_0^r (t - \xi(t)) dt.$$

Then

$$D(r) = 2 \exp\left(-\int_0^r \xi(t) dt\right), \quad 0 < D(r) < 2 \quad (r > 0), \quad (43)$$

and $D(r) \leq 2e^{-\tau r^2/2}$.

Proof. Write

$$F_0(r, s) := \frac{r}{s} - 1 + \frac{r^2 - s^2}{2} \quad (r, s > 0).$$

Its derivative with respect to s is $-r/s^2 - s < 0$. On the lower line $s = \tau r$, using $\tau^2 + \tau = 1$, one has

$$F_0(r, \tau r) = \tau + \frac{\tau r^2}{2} > \tau.$$

On the upper line $s = r$, $F_0(r, r) = 0 < 1$. These are lower and upper barriers. Another upper barrier is $q(r) := \tau r + r^3/8$: since $q(r)/r \geq \tau$,

$$F_0(r, q(r)) \leq \tau + \frac{\tau r^2}{2} \leq \tau + \frac{3r^2}{8} = q'(r),$$

where $\tau < 3/4$ was used.

For $0 < \epsilon < 1$, solve the ordinary initial-value problem from $\xi_\epsilon(\epsilon) = \tau\epsilon$. The barriers keep the solution positive and prevent finite-time escape on compact intervals. Its auxiliary quantity $D_\epsilon := 2 + \xi_\epsilon^2 - r^2$ satisfies

$$D'_\epsilon = -\xi_\epsilon D_\epsilon, \quad D_\epsilon(\epsilon) = 2 - \tau\epsilon^2 > 0.$$

It follows that $0 < D_\epsilon < 2$ and

$$0 < \xi'_\epsilon = r/\xi_\epsilon - D_\epsilon/2 < 1/\tau < 2.$$

Extend $\xi_\epsilon(r) := \tau r$ on $[0, \epsilon]$. The barriers and these Lipschitz bounds give a subsequential limit on every compact interval as $\epsilon \downarrow 0$. On intervals bounded away from zero, passing to the integral form of the differential equation shows that the limit solves Eq. (42). The barriers pass to the limit and give continuity at zero and the stated slope there.

For uniqueness, two positive solutions have difference w satisfying $w' = c(r)w$ with $c(r) < 0$, by the mean value theorem in s . For $r \geq \epsilon$, $|w(r)| \leq |w(\epsilon)|$. Continuity at zero and then $\epsilon \downarrow 0$ show $w(r) = 0$. This proves uniqueness even without separately imposing the initial slope.

For the limiting solution, differentiation gives $D' = -\xi D$ and $D(0) = 2$, proving Eq. (43). The lower barrier on ξ gives the exponential upper bound. Since $D < 2$ and $\xi < r$, the displayed formula for ξ' proves strict positivity; its upper bound follows from $\xi \geq \tau r$ and $D > 0$. The lower barrier also makes the range of ξ unbounded, completing the proof. $\square$

4.6 The comparison profile and exact constant

Let φ and Φ be the standard normal density and distribution function. Put

$$B(r) := \varphi(0)e^{-I(r)}, \quad P(r) := \Phi(-r) + (r + \xi(r))B(r). \quad (44)$$

Define $\mathcal{V} : \mathbb{R} \rightarrow \mathbb{R}$ by

$$\mathcal{V}(x) := \begin{cases} \varphi(x) + x\Phi(x), & x \leq 0, \\ xP(r) + B(r), & x > 0, \end{cases} \quad r := \xi^{-1}(x).$$

The following identities establish the regularity and normalization needed to turn this profile into a majorant of the positive part.

Lemma 4.13 (Profile identities and normalization). *The function $\mathcal{V}$ is twice continuously differentiable and convex. Its second derivative increases strictly on $(-\infty, 0)$ and decreases strictly on $(0, \infty)$. There is a finite positive limit $P_\infty := \lim_{r \rightarrow \infty} P(r)$. Define*

$$\Gamma_B := \frac{\varphi(0)}{P_\infty} = \lim_{r \rightarrow \infty} \frac{e^{I(r)}}{r + \xi(r)}. \quad (45)$$

Then

$$\mathcal{V}(x) \geq P_\infty x_+ \quad (x \in \mathbb{R}). \quad (46)$$

For $x = \xi(r) > 0$,

$$\mathcal{V}'(x) = P(r), \quad (47)$$

$$\mathcal{V}(x) - x\mathcal{V}'(x) = B(r), \quad (48)$$

$$\mathcal{V}''(x) = \frac{2B(r)}{2 + x(x + r)}. \quad (49)$$

Proof. Eq. (43) and the definition of I give

$$B(r) = \frac{2\varphi(r)}{D(r)}, \quad B'(r) = -(r - \xi(r))B(r).$$

Differentiation of P gives

$$P'(r) = B(r) \frac{r - \xi(r)}{\xi(r)} > 0. \quad (50)$$

For clarity, the cancellation is

$$P'(r) = -\varphi(r) + B(r)[1 + \xi'(r) - (r^2 - \xi(r)^2)] = B(r)(r/\xi(r) - 1).$$

Here $\varphi(r) = B(r)D(r)/2$ and $\xi' = r/\xi - D/2$ were used.

Differentiating $\xi(r)P(r) + B(r)$ now gives exactly $\xi'(r)P(r)$, proving Eq. (47) and Eq. (48). The factorization

$$2r - xD(r) = (r - x)(2 + x(x + r)), \quad x := \xi(r),$$

together with $\xi' = (2r - xD)/(2x)$ proves Eq. (49).

At zero, $B \rightarrow \varphi(0)$, $P \rightarrow 1/2$, and the second derivative in Eq. (49) tends to $\varphi(0)$. These agree with the value and the first two derivatives of $\varphi(x) + x\Phi(x)$ from the left. Thus $\mathcal{V}$ is C^2 across zero. On the negative half-line $\mathcal{V}'' = \varphi$, which increases there. On the positive half-line B decreases strictly and the denominator in Eq. (49) increases strictly, proving the asserted decrease and convexity.

For $r \geq 2$, $D > 0$ implies $\xi(r) > \sqrt{r^2 - 2} \geq r/\sqrt{2}$. Also

$$r - \xi(r) = \frac{2 - D(r)}{r + \xi(r)} \leq \frac{2}{r}.$$

Since $B(r) \leq \varphi(0)$, Eq. (50) is bounded above by $2\sqrt{2}\varphi(0)/r^2$. Thus the increasing function P has a finite limit, and that limit is positive because $P(0) = 1/2$. As $\Phi(-r) \rightarrow 0$, Eq. (44) proves the second equality in Eq. (45).

The negative branch of $\mathcal{V}$ is positive, for example from its integral representation $\mathbb{E}[(x + Z)_+]$. For $x > 0$,

$$\frac{d}{dx} \frac{\mathcal{V}(x)}{x} = -\frac{B(\xi^{-1}(x))}{x^2} < 0,$$

and $\mathcal{V}(x)/x \rightarrow P_\infty$, because B is bounded. This proves Eq. (46) on the positive half-line as well. $\square$

We verify that the comparison profile also controls every downward jump.

Lemma 4.14 (Global differential inequality). *For every real x and every $h \geq 0$,*

$$\mathcal{V}(x - h) - \mathcal{V}(x) + h\mathcal{V}'(x) \leq \frac{h^2}{2}[\mathcal{V}(x) - x\mathcal{V}'(x)]. \quad (51)$$

Proof. The case $h = 0$ is immediate. If $x \leq 0$, Taylor's integral remainder and monotonicity of $\mathcal{V}''$ on $(-\infty, 0]$ give a remainder at most $h^2\varphi(x)/2$. On that branch $\mathcal{V}(x) - x\mathcal{V}'(x) = \varphi(x)$, proving the claim.

Suppose $x > 0$ and set $r := \xi^{-1}(x)$, $b := B(r)$. For $t \leq x$, put

$$G(t) := \mathcal{V}(t) - \mathcal{V}(x) - (t - x)\mathcal{V}'(x) - \frac{b}{2}(t - x)^2.$$

We will prove $G(t) \leq 0$. By construction $G(x) = G'(x) = 0$. The identities in Lemma 4.13 also give

$$G(-r) = G'(-r) = 0.$$

Indeed, $\mathcal{V}'(-r) = \Phi(-r)$, $P(r) - \Phi(-r) = (x+r)b$, and $\mathcal{V}(-r) = \varphi(r) - r\Phi(-r)$. Using $2\varphi(r) = bD(r)$, the remaining value identity reduces to

$$D(r) - 2 + 2r(x+r) - (x+r)^2 = 0,$$

which follows from $D(r) = 2 + x^2 - r^2$.

Now $G''(t) = \mathcal{V}''(t) - b$. One has $0 < b < \varphi(0)$, $\mathcal{V}''(-r) = \varphi(r) = bD(r)/2 < b$, and $\mathcal{V}''(x) = 2b/(2 + x(x+r)) < b$. The strict increase and decrease of $\mathcal{V}''$ therefore give unique numbers $\alpha \in (-r, 0)$ and $\beta \in (0, x)$ such that G'' is negative on $(-\infty, \alpha)$, positive on (α, β) , and negative on $(\beta, x]$.

On $(-\infty, -r)$, $G'(t) > G'(-r) = 0$, so $G(t) < G(-r) = 0$. On $(-r, \alpha]$, $G' < 0$, whereas on $[\beta, x)$, $G' > 0$ because $G'(x) = 0$ and $G'' < 0$ there. The strictly increasing derivative on (α, β) consequently has exactly one zero. Thus on $[-r, x]$ the function G first decreases and then increases, and its endpoint values are both zero. This proves $G \leq 0$ throughout $(-\infty, x]$. Taking $t := x - h$ proves Eq. (51). $\square$

The two tangencies at $-r$ and x prove the inequality for every real state and every nonnegative jump size.

Theorem 4.15 (Direct positive-part bound). *Let L be a bounded martingale with $L_0 = 0$, obtained from finitely many compensated simple counting processes with predictable intensities and no simultaneous jumps. Assume its jumps are nonpositive and its predictable quadratic variation is continuous. Suppose that for a deterministic $V > 0$, $\langle L \rangle_T \leq V$ on every path. Then*

$$\mathbb{E}[(L_T)_+] \leq \Gamma_B \sqrt{V}.$$

In particular, the theorem applies to the finite-stage embedding of the entropy-drift edge score.

Proof. Set $\mathcal{U}(t, x) := \sqrt{t}\mathcal{V}(x/\sqrt{t})/P_\infty$ for $t > 0$. By Lemma 4.13, this function is $C^{1,2}$ and majorizes x_+ . Writing $z := x/\sqrt{t}$, differentiation gives

$$\mathcal{U}_t(t, x) = \frac{\mathcal{V}(z) - z\mathcal{V}'(z)}{2P_\infty\sqrt{t}}.$$

Apply Eq. (51) at state z with jump $h/\sqrt{t}$, and multiply by $\sqrt{t}/P_\infty$. This yields

$$\mathcal{U}(t, x-h) - \mathcal{U}(t, x) + h\mathcal{U}_x(t, x) \leq h^2\mathcal{U}_t(t, x) \quad (h \geq 0).$$

Lemma 4.10, with $g(x) := x_+$, now gives

$$\mathbb{E}[(L_T)_+] \leq \mathcal{U}(V, 0) = \Gamma_B \sqrt{V}.$$

$\square$

The scalar curve and Lemma 4.14 already prove an exact upper bound with coefficient Γ_B . To replace it by a simple explicit rational, it remains to bound only one scalar integral.

Lemma 4.16 (An analytic enclosure of the infinite tail). *For $R > \sqrt{2}$, let $b_R := \sqrt{R^2 - 2}$ and*

$$G_R := \frac{\exp(I(R) - (R^2 - Rb_R - 1)/2)}{R + b_R}.$$

Then

$$G_R \exp(-D(R)/(2(R^2 - 2))) \leq \Gamma_B \leq G_R.$$

Proof. For $r \geq R$, put $b(r) := \sqrt{r^2 - 2}$. From $D > 0$,

$$\xi(r) > b(r), \quad \xi(r) - b(r) = \frac{D(r)}{\xi(r) + b(r)}.$$

The identity $D' = -\xi D$ gives $D(r) \leq D(R)e^{-b_R(r-R)}$. Therefore

$$0 \leq E_R := \int_R^\infty (\xi(r) - b(r))dr \leq \frac{D(R)}{2b_R^2}.$$

An antiderivative of $r - b(r)$ is

$$A_0(r) := \frac{r^2 - rb(r)}{2} + \log(r + b(r)).$$

Also $r^2 - rb(r) \rightarrow 1$ and $r + \xi(r) \sim r + b(r) \sim 2r$. Substitution into Eq. (45) gives the exact identity $\Gamma_B = G_R e^{-E_R}$, proving the lemma. $\square$

The following finite calculation uses rational interval endpoints. All rounding operations and truncation parameters are specified below.

Set $r_0 := 1/64$, $R := 6$, and mesh $h := 2^{-14}$. Put $k := \lfloor \sqrt{5 \cdot 10^{80}} \rfloor$ and set $\tau_- := (k/10^{40} - 1)/2$ and $\tau_+ := ((k+1)/10^{40} - 1)/2$. These are outward rational bounds on τ . Lemma 4.12 initializes the intervals

$$\begin{aligned} \xi(r_0) &\in [\tau_- r_0, \tau_+ r_0 + r_0^3/8], \\ I(r_0) &\in [(1 - \tau_+)r_0^2/2 - r_0^4/32, (1 - \tau_-)r_0^2/2]. \end{aligned}$$

For $r > 0$, define the second-order Taylor map

$$T_h(r, s) := s + hF_0(r, s) + \frac{h^2}{2}[(F_0)_r(r, s) + (F_0)_s(r, s)F_0(r, s)].$$

The true curve has $r/2 \leq \xi(r) \leq r$ and $0 < \xi' < 2$. Differentiating the equation gives

$$|\xi''(r)| \leq \frac{10}{r} + 3r, \quad |\xi'''(r)| \leq \frac{120}{r^2} + 27 + 3r^2.$$

To verify these bounds directly, use

$$(F_0)_r = 1/s + r, \quad (F_0)_s = -r/s^2 - s, \quad (F_0)_{rr} = 1, \quad (F_0)_{rs} = -1/s^2, \quad (F_0)_{ss} = 2r/s^3 - 1$$

in the identities for the second and third derivatives, with $s \geq r/2$ and $|\xi'| < 2$.

The map $s \mapsto T_h(r, s)$ is increasing on $[r/2, r]$ on the entire mesh. An explicit lower bound for its derivative is

$$1 - h(4/r + r) - \frac{h^2}{2}[100/r^2 + 100(1 + r^2)] > 0.$$

Indeed, on this strip $0 \leq F_0 \leq 1 + r^2/2$, and the absolute value of $\partial_s[(F_0)_r + (F_0)_s F_0]$ is at most $36/r^2 + 17 + 3r^2/2$, smaller than the displayed conservative bound. Positivity follows by substituting $r \geq r_0$ and $r \leq R$ in this lower bound.

If $l \leq \xi(r) \leq u$, the next enclosure is consequently

$$l' := T_h(r, l) - e_r, \quad u' := T_h(r, u) + e_r, \quad e_r := \frac{h^3}{6}[120/r^2 + 27 + 3(r+h)^2].$$

Round these endpoints outward to multiples of 10^{-32} , then intersect with the rounded analytic barriers $\tau_-(r+h) \leq \xi(r+h) \leq \min\{r+h, \tau_+(r+h) + (r+h)^3/8\}$. Initialize the four endpoints at r_0 with the same outward rounding. For the integral, the trapezoidal rule has error at most

$$e_{I,r} := \frac{h^3}{12}[10/r + 3(r+h)].$$

Thus if $I(r) \in [i_-, i_+]$, valid updated endpoints are

$$\begin{aligned} i'_- &:= i_- + \frac{h}{2}(2r+h-u-u') - e_{I,r}, \\ i'_+ &:= i_+ + \frac{h}{2}(2r+h-l-l') + e_{I,r}. \end{aligned}$$

They too are rounded outward. This proves by induction that every mesh step encloses the exact solution and its integral; it is not a grid sampling assertion.

There are exactly 98048 steps. This recurrence gives, with endpoints rounded further outward here for display,

$$\begin{aligned} 1.67911709307509 &< I(6) < 1.67911767026181, \\ 0.44989261230112 &< \Gamma_B < 0.44989287305358. \end{aligned}$$

For exponentials at a rational $x \geq 0$, sum the Taylor series through degree $N := 100$ and bound its positive tail by

$$\frac{x^{N+1}}{(N+1)!} \frac{1}{1 - x/(N+2)} \quad (N+2 > x).$$

Negative exponents are handled by reciprocating positive enclosures. For $x \geq 0$, enclose $\sqrt{x}$ by $[k/10^{40}, (k+1)/10^{40}]$, where $k := \lfloor \sqrt{\lfloor 10^{80}x \rfloor} \rfloor$. Exponential and logarithmic endpoints are rounded outward to multiples of 10^{-32} . Finally,

$$\log 2 = 2 \sum_{j \geq 0} \frac{(1/3)^{2j+1}}{2j+1}$$

is enclosed using the terms $j = 0, \dots, 79$ and the tail bound $2(1/3)^{161}/(161(1 - 1/9))$. Together with Lemma 4.16, these operations certify the exact strict inequalities

$$\Gamma_B < \frac{44989288}{100000000}, \quad \frac{44989288}{100000000} \sqrt{\log 2} < \frac{3745604}{100000000}.$$

The interval proof covers $(0, r_0]$ by analytic barriers, $[r_0, 6]$ by validated integration, and $[6, \infty)$ by an analytic tail estimate. The jump inequality itself has already been proved for every real state and every nonnegative jump size, independently of this computation.

4.7 Consequences and Gaussian comparison

Proposition 4.17 (Strengthened entropy-score estimate). *For every $n \geq 1$, $0 < a < 1$, $u > 1$, and nonnegative normalized density f , put $\ell := \log u$ and $y := u \Pr_\mu[T_a f \geq u]$. Then*

$$y\sqrt{\ell} \leq \Gamma_B \sqrt{(\kappa_a + 1/2)\rho_a} + \sqrt{y}. \quad (52)$$

In particular,

$$C_*(a) \leq \Gamma_B \sqrt{(\kappa_a + 1/2)\rho_a} + 1. \quad (53)$$

Proof. The doubly capped entropy score satisfies

$$y\ell \leq \mathbb{E}_Q[S_+] + \sqrt{y\ell}, \quad \sum_i v_i^2 \leq \rho_a \ell \quad \text{pathwise.}$$

Its deterministic-stage embedding is a bounded downward-jump martingale with the same terminal score and bracket budget $V := (\kappa_a + 1/2)\rho_a \ell$. These are Lemma 4.9 and Lemma 4.11. Theorem 4.15 gives $\mathbb{E}_Q[S_+] \leq \Gamma_B \sqrt{V}$. Substitute and divide by $\sqrt{\ell} > 0$ to obtain Eq. (52). Since $y \leq 1$ by Markov's inequality, taking the supremum over all n, u, f proves Eq. (53). $\square$

For large entropy times we also use a Gaussian comparison.

The following bounded-jump estimate will cover the unbounded time range in Section 4.10.

Lemma 4.18 (Gaussian smoothing with a bounded-jump correction). *Let L be a bounded martingale starting at zero, obtained from finitely many compensated counting processes on a deterministic finite horizon. Suppose its jumps lie in $[-b, 0]$ and its continuous predictable quadratic variation is at most a deterministic $V > 0$. Then, for every $\delta > 0$,*

$$\mathbb{E}[(L_n)_+] \leq \varphi(0)\sqrt{V + \delta} + \frac{b\varphi(1)}{6} \log(1 + V/\delta). \quad (54)$$

The lemma applies in particular to the deterministic-stage score embedding.

Proof. Let $\mathcal{G}(t, x) := \mathbb{E}[(x + \sqrt{t}Z)_+]$ for a standard normal Z . Then $\mathcal{G} \geq x_+$, $\mathcal{G}_t = \mathcal{G}_{xx}/2$, and

$$\sup_x |\mathcal{G}_{xxx}(t, x)| = \varphi(1)/t.$$

The last identity follows from $\mathcal{G}_{xxx}(t, x) = -x\varphi(x/\sqrt{t})/t^{3/2}$ and $\sup_z |z|\varphi(z) = \varphi(1)$. Taylor's formula yields, for $0 \leq h \leq b$,

$$\mathcal{G}(t, x - h) - \mathcal{G}(t, x) + h\mathcal{G}_x(t, x) \leq \frac{h^2}{2}\mathcal{G}_{xx}(t, x) + \frac{bh^2\varphi(1)}{6t}.$$

Use the remaining bracket time $t_s := V + \delta - \langle L \rangle_s \geq \delta$. The quadratic term cancels the bracket-clock contribution. Integrating the remaining compensator term gives at most

$$\frac{b\varphi(1)}{6} \int_0^{\langle L \rangle_n} \frac{dv}{V + \delta - v} \leq \frac{b\varphi(1)}{6} \log(1 + V/\delta).$$

Boundedness and $t_s \geq \delta$ justify the expectation calculation. At the terminal time $\mathcal{G}$ majorizes the positive part, and initially $\mathcal{G}(V + \delta, 0) = \varphi(0)\sqrt{V + \delta}$, proving the claim. Grouped simultaneous jumps can also be used, provided the bound b applies to their total downward size. $\square$

4.8 The joint entropy score and verification principle

Put

$$c_j := \frac{4541}{12500} = 0.36328. \quad (55)$$

The improvement below keeps the exact entropy constraint in certified jump intervals and verifies the spatial state and jump size jointly. It also retains the two different time-endpoint requirements instead of charging their maximum throughout each time interval.

Theorem 4.19 (Joint entropy-cost upper bound). *For every integer $n \geq 1$, $0 < a < 1$, $u > 1$, and nonnegative f with $\mathbb{E}_\mu[f] = 1$, set $\ell := \log u$ and $y := u \Pr_\mu[T_a f \geq u]$. Then*

$$y\sqrt{\ell} \leq c_j \sqrt{\kappa_a + 1} + \sqrt{y}. \quad (56)$$

Consequently, for every $0 < a < 1$,

$$C_*(a) \leq c_j \sqrt{\kappa_a + 1} + 1, \quad \limsup_{a \uparrow 1} \frac{C_*(a)}{\sqrt{\kappa_a}} \leq c_j < \frac{9247}{25000}. \quad (57)$$

These bounds are uniform before taking any supremum over the dimension, threshold, or input. They do not assert existence of the normalized limit.

Lemma 4.9 gives the doubly capped score estimates

$$y\ell \leq \mathbb{E}_Q[S_+] + \sqrt{y\ell}, \quad D_n := \sum_i e(s_i, r_i) \leq \ell, \quad \alpha_i := |v_i| = e(s_i, r_i)/s_i, \quad (58)$$

where zero coefficients are used at $s_i = 0$, $0 \leq r_i \leq s_i \leq a$, and

$$e(s, r) := s \operatorname{arctanh}(r) + \frac{1}{2} \log(1 - r^2), \quad d(s) = e(s, s), \quad d(1) = \log 2.$$

In particular, $\alpha_i \leq d(s_i)/s_i$, $\sum_i \alpha_i^2 \leq \rho_a \ell$, and $\rho_a := d(a)/a^2 \leq \log 2$. The non-strict edge comparison and double barrier were proved in Sections 4.3 and 4.4, respectively.

Write

$$K := \kappa_a + 1, \quad T := \ell/K, \quad L := S/K.$$

In the deterministic-stage embedding of Lemma 4.11, conditional on the latent vector and the observed prefix, a nonzero stage has parameters

$$\alpha := e_i/s_i, \quad \beta \in \{\kappa_a \alpha, \alpha/\kappa_a\}.$$

Before the stage jump, the embedding has intensity $\lambda(z) := \alpha/(\beta + \alpha z)$ at elapsed time $z \in [0, 1]$. Consume entropy time at rate e_i/K throughout the stage, including after the jump. The normalized jump size h then satisfies

$$h = \frac{\beta + \alpha z}{K} \leq \frac{d(s_i)}{s_i}, \quad \lambda(z) s_i h = e_i/K.$$

Zero-coefficient stages leave the score and the entropy clock unchanged. The embedding lemma already proves that this construction preserves the full adaptive joint law.

Set

$$b := \frac{693147181}{1000000000} > \log 2.$$

The logarithmic enclosure in Appendix A.1 certifies the last inequality.

Lemma 4.20 (Certified inverse-entropy intervals). *Let $v > 0$, $0 < s_0 \leq 1$, and $p > 0$ satisfy*

$$d(s_0) \leq vs_0, \quad ps_0 \geq v. \quad (59)$$

For every $0 < s \leq 1$ and $0 \leq h \leq \min\{v, d(s)/s\}$,

$$\frac{h}{s} \leq p, \quad sh \geq h^2/p.$$

Proof. The positive series

$$d(s) = \sum_{j \geq 1} \frac{s^{2j}}{(2j)(2j-1)}$$

shows that $d(s)/s^2$ increases on $(0, 1]$. If $s \geq s_0$, then $h/s \leq v/s_0 \leq p$. If $s < s_0$, then

$$\frac{h}{s} \leq \frac{d(s)}{s^2} \leq \frac{d(s_0)}{s_0^2} \leq \frac{v}{s_0} \leq p.$$

Multiplication gives the second inequality; at $h = 0$ it is immediate. This proves the entire bias interval from each rational test in Eq. (59). $\square$

For a mesh denominator N , set $\Delta := 1/N$, $J := \lfloor bN \rfloor$, and

$$v_k := \min\{(k+1)\Delta, b\}, \quad 0 \leq k \leq J.$$

Appendix A.1 constructs the rational pairs $s_k := b_k/10^{12}$ and $p_k := P_k/10^8$ satisfying Eq. (59) with $v := v_k$. The small-jump interval is $[0, \Delta]$; the other intervals are $[k\Delta, v_k]$ for $1 \leq k \leq J$. Together they cover $[0, b]$. These are certificate meshes, not restrictions on the Boolean dimension.

Lemma 4.21 (Entropy-clock verification with interval costs). *Let U be continuous on $[0, T_0] \times \mathbb{R}$, continuously differentiable in space with locally Lipschitz spatial derivative, and piecewise continuously differentiable in time on a finite partition. Assume the functions and derivatives have at most linear growth in space. Suppose $U(0, x) \geq x_+$, $U_t \geq 0$, and, throughout each time interval including both one-sided endpoints,*

$$p(h)[U(t, x-h) - U(t, x) + hU_x(t, x)] \leq h^2U_t(t, x) \quad (60)$$

for all real x and all h in jump interval k . Then

$$\mathbb{E}_Q[S_+] \leq KU(T, 0), \quad 0 < T \leq T_0. \quad (61)$$

Proof. Time monotonicity gives $U(t, x) \geq x_+$. Write $R := U(t, x-h) - U(t, x) + hU_x(t, x)$. At a nonzero stage, Lemma 4.20 gives $h^2 \leq p_k sh$. Since $U_t \geq 0$, Eq. (60) yields

$$p_k R \leq h^2 U_t \leq p_k sh U_t.$$

Thus $R \leq sh U_t$, with no division by h and no sign condition on R . Before a stage jump, the compensator contribution to $U(t, L)$ is therefore

$$\lambda[R - sh U_t] \leq 0.$$

After a jump, only entropy time continues changing, with contribution $-(e_i/K)U_t \leq 0$. Zero stages contribute nothing.

For each fixed finite input, the horizon is the deterministic integer n , there are at most n jumps, all intensities are at most κ_a , and the states and coefficients are bounded. Between jumps the paths are absolutely continuous. The ordinary chain rule applies; the finitely many time knots cause no extra variation term because U is continuous. If the clock remains at a knot, its rate is zero and the score is constant. Taking expectations is consequently justified directly. At the terminal time $T - D_n/K \geq 0$, so

$$\mathbb{E}_Q[U(T - D_n/K, L_n)] \leq U(T, 0).$$

The positive-part majorant proves Eq. (61). $\square$

4.9 Spline construction and polynomial certification

Both witnesses use the coefficient and time scale

$$S_0 := 10^{12}, \quad G_0 := 10^8.$$

The integer coefficient C_{mi} represents $p_i(t_m) := C_{mi}/S_0$ at the center $(i + 1/2)\Delta$, and the integer A_m represents $t_m := A_m/S_0$. The initial coefficients are exactly $((i + 1/2)\Delta)_+$. All coefficients outside the stored interval are defined for every time by the exact affine tails: zero on the left and $(i + 1/2)\Delta$ on the right. Between time knots interpolate p_i affinely. For $x = (i + r)\Delta$, $0 \leq r \leq 1$, define

$$P(t, x) := \frac{(1 - r)^2}{2} p_{i-1}(t) + (1/2 + r - r^2) p_i(t) + \frac{r^2}{2} p_{i+1}(t). \quad (62)$$

The weights are nonnegative, sum to one, and have weighted center x . Thus $P(0, x) \geq x_+$. Adjacent cells agree in value and first derivative. On the interior of cell i ,

$$P_{xx}(t, x) = N^2[p_{i-1}(t) - 2p_i(t) + p_{i+1}(t)].$$

No convexity of later coefficient rows is assumed.

On time interval m , put $a := A_{m+1} - A_m$ and $\tau := (t - t_m)/(t_{m+1} - t_m)$. The finite offset procedure specified below constructs two nonnegative integers δ_m^- and δ_m^+ , with even sum, and defines

$$\begin{aligned} E(t) &:= E(t_m) + \frac{\delta_m^- \tau + (\delta_m^+ - \delta_m^-) \tau^2 / 2}{S_0}, \\ E(t_{m+1}) - E(t_m) &= \frac{\delta_m^- + \delta_m^+}{2S_0}, \quad E(0) := 0. \end{aligned} \quad (63)$$

Set $U := P + E$. Its time derivative is

$$U_t = P_t + \frac{(1 - \tau)\delta_m^- + \tau\delta_m^+}{a}. \quad (64)$$

In particular, E is nondecreasing. The spatial remainder in Eq. (60) is independent of E and affine in t , while U_t is also affine in t . It therefore suffices to prove Eq. (60) at the two time endpoints, using their respective δ_m^- and δ_m^+ . Charging their average in Eq. (63), rather than their maximum throughout the interval, is valid at every intermediate time.

We now express the state and jump constraints jointly as polynomial inequalities. Fix a time interval and define the raw temporal differences $D_i := C_{m+1,i} - C_{mi}$. For any integer row C , put

$$A_i^C := C_{i-1} + C_i, \quad B_i^C := 2(C_i - C_{i-1}), \quad Q_i^C := C_{i-1} - 2C_i + C_{i+1}.$$

Thus the doubled raw spatial spline is $A_i^C + B_i^C r + Q_i^C r^2$. Define

$$D_i^{(2)}(r) := A_i^D + B_i^D r + Q_i^D r^2.$$

For $k \geq 1$, write $h := (k+v)\Delta$, $0 \leq v \leq 1$. Split the square $0 \leq r, v \leq 1$ into the two closed triangles $v \leq r$ and $r \leq v$. In the first triangle let $j := i - k$ and $\epsilon := 0$; in the second let $j := i - k - 1$ and $\epsilon := 1$. Then the doubled raw remainder is the quadratic

$$\begin{aligned} R_{i,k}^C(r, v) &:= A_j^C + B_j^C \epsilon + Q_j^C \epsilon^2 - A_i^C + k B_i^C \\ &\quad + (B_j^C + 2Q_j^C \epsilon - B_i^C + 2k Q_i^C) r \\ &\quad + (-B_j^C - 2Q_j^C \epsilon + B_i^C) v \\ &\quad + (Q_j^C - Q_i^C) r^2 + (-2Q_j^C + 2Q_i^C) r v + Q_j^C v^2. \end{aligned}$$

Indeed, this is exactly the expansion of

$$(A_j^C + B_j^C(r - v + \epsilon) + Q_j^C(r - v + \epsilon)^2) - (A_i^C + B_i^C r + Q_i^C r^2) + (k + v)(B_i^C + 2Q_i^C r).$$

Write $p_k := P_k/G_0$, with integer P_k . At time endpoint $e \in \{-, +\}$, let C be the corresponding coefficient row and let $\delta := \delta_m^e$. Clearing denominators in Eq. (60) gives precisely

$$S_0 G_0 (k + v)^2 [D_i^{(2)}(r) + 2\delta] - a N^2 P_k R_{i,k}^C(r, v) \geq 0. \quad (65)$$

Here is a finite sufficient test over each entire triangle. Introduce $\lambda_0, \lambda_1, \lambda_2 \geq 0$ with sum one. On the first triangle take $r := \lambda_1 + \lambda_2$, $v := \lambda_2$; on the second take $r := \lambda_2$, $v := \lambda_1 + \lambda_2$. Homogenize all terms in Eq. (65) to degree four by powers of $\lambda_0 + \lambda_1 + \lambda_2$. If all fifteen coefficients are nonnegative, the polynomial is nonnegative everywhere on the triangle. This is the elementary nonnegative-coefficient form of the Bernstein test, not an evaluation on a grid.

For clarity, let $H_2 := (k(\lambda_0 + \lambda_1 + \lambda_2) + v)^2$, let D_2 and R_2 be the degree-two homogenizations of $D_i^{(2)}$ and $R_{i,k}^C$, and let $B_2 := (\lambda_0 + \lambda_1 + \lambda_2)^2$. Form

$$D_4 := H_2 D_2, \quad R_4 := R_2 B_2, \quad E_4 := H_2 B_2.$$

Every coefficient of E_4 is strictly positive because $k \geq 1$. For each degree-four monomial, with coefficients denoted by brackets, the sufficient integer condition is

$$\delta \geq \max\{0, \lceil \frac{a N^2 P_k [R_4] - S_0 G_0 [D_4]}{2 S_0 G_0 [E_4]} \rceil\}. \quad (66)$$

Both triangles and both time endpoints are checked. In the last jump interval this checks a possibly larger triangle than is necessary; the actual partial interval is a subset.

The interval containing zero requires a separate treatment that includes arbitrarily small jumps. For $0 \leq h \leq \Delta$, write $h := v\Delta$. When $v \leq r$, the jump stays in one cell. Canceling h^2 reduces Eq. (60) to

$$S_0 G_0 [D_i^{(2)}(r) + 2\delta] - a N^2 P_0 Q_i^C \geq 0. \quad (67)$$

The three quadratic Bernstein coefficients of $D_i^{(2)}$ on $[0, 1]$ are

$$A_i^D, \quad A_i^D + B_i^D/2 = 2D_i, \quad A_i^D + B_i^D + Q_i^D = D_i + D_{i+1}. \quad (68)$$

Testing Eq. (67) at these three coefficients is sufficient for every r .

When $r \leq v$, put $r := vz$ with $v, z \in [0, 1]$. Taylor's integral remainder, or cancellation using the two C^1 knot identities, gives

$$\frac{U(t, x - h) - U(t, x) + hU_x(t, x)}{h^2} = \frac{N^2}{2S_0} [Q_i^C z(2 - z) + Q_{i-1}^C (1 - z)^2]$$

at a time endpoint. Thus the required polynomial is

$$S_0 G_0 [D_i^{(2)}(vz) + 2\delta] - aN^2 P_0 [Q_{i-1}^C + (2Q_i^C - 2Q_{i-1}^C)z + (Q_{i-1}^C - Q_i^C)z^2].$$

It has degree at most two in each of v, z . Its nine tensor Bernstein coefficients must be nonnegative. Explicitly, multiply a coefficient by four and index it by $j, k \in \{0, 1, 2\}$. The test is

$$\begin{aligned} S_0 G_0 [4A_i^D + 8\delta + B_i^D jk + 4Q_i^D \mathbf{1}_{\{j=k=2\}}] \\ - aN^2 P_0 [4Q_{i-1}^C + 2k(2Q_i^C - 2Q_{i-1}^C) + 4(Q_{i-1}^C - Q_i^C) \mathbf{1}_{\{k=2\}}] \geq 0. \end{aligned} \quad (69)$$

These tests include the limiting cases $v = 0$, so no lower cutoff on jump size is imposed. Nonnegativity of U_t is separately verified by adding 2δ to all three coefficients in Eq. (68).

The following coarser bound is a sufficient preliminary test that reduces the number of arithmetic operations. It does not replace a failed polynomial test by a numerical estimate.

On a time interval let $\bar{q}_i$ be the larger of the two endpoint curvatures of P , and put $Q_i := \max\{0, \bar{q}_{i-1}, \bar{q}_i, \bar{q}_{i+1}\}$. For a real $x \in [i\Delta, (i+1)\Delta]$ and $h \in [k\Delta, v_k]$, $k \geq 1$, Taylor's integral formula gives

$$\frac{U(t, x - h) - U(t, x) + hU_x(t, x)}{h^2} \leq \frac{1}{2k^2} \sum_{j=1}^k (2k - 2j + 1) Q_{i-j}.$$

To see this, compare $x - zh$ with $i\Delta - zk\Delta$; their distance is at most Δ . For $z \in [(j-1)/k, j/k]$, the reference point lies in cell $i-j$, so the true curvature is at most Q_{i-j} . Integrating $1 - z$ gives the displayed weights. For $h \leq \Delta$, the corresponding bound is $\max\{0, \bar{q}_{i-1}, \bar{q}_i\}/2$.

In integer units put $q_i^\# := N^2 Q_i^C$, form the same endpoint maxima and three-cell maxima, and define

$$W_{ik}^\# := \sum_{j=1}^k (2k - 2j + 1) Q_{i-j}^\#, \quad d_i := \min\{D_{i-1}, D_i, D_{i+1}\}.$$

With $\delta_* := \min\{\delta_m^-, \delta_m^+\}$, a sufficient preliminary test is

$$aP_k W_{ik}^\# \leq 2S_0 G_0 k^2 (d_i + \delta_*).$$

The analogous small-jump test has numerator $aP_0 \max\{0, \bar{q}_{i-1}^\#, \bar{q}_i^\#\}$ and denominator factor $2S_0 G_0$.

Visit time intervals in increasing order, starting each with $(\delta_m^-, \delta_m^+) := (0, 0)$. Within each interval, visit spatial cells in increasing order. At each cell first enforce the three time-derivative tests in Eq. (68) for both endpoint integers, and then visit jump bins $k = 0, 1, \dots, J$ in this order. Each preliminary test uses the current $\delta_* := \min\{\delta_m^-, \delta_m^+\}$. If it passes, no update is made. If it fails at $k = 0$, enforce all three same-cell tests and all nine crossing tests in Eqs. (67) and (69), for both time endpoints. If it fails at $k \geq 1$, enforce all fifteen inequalities in Eq. (66), on both triangles and at both time endpoints. To enforce a rational lower bound $\delta \geq b$, replace δ by $\max\{\delta, 0, \lceil b \rceil\}$; thus every update is the smallest permissible integer increase. Raising an endpoint integer cannot invalidate an earlier test. After all cells, if $\delta_m^- + \delta_m^+$ is odd, raise δ_m^+ by one, and integrate using Eq. (63). This specifies the offsets exactly.

The 8000 stored coefficients have original indices $-4000, \dots, 3999$. Pad them by $J + 5$ exact affine-tail coefficients on each side, obtaining $\tilde{n} := 8000 + 2(J + 5)$ entries. Padded indices are zero-based: padded index j represents original index $j - 4000 - (J + 5)$. Visit every cell with padded index $j = J + 2, \dots, \tilde{n} - 2$. The omitted far-left cells are identically in the zero tail. Beyond the right end the whole segment $[x - b, x]$ lies in the right affine tail. In both omitted regions the spatial remainder vanishes and $U_t = E' \geq 0$. All joins between stored data and affine tails are inside the checked region. Thus the finite test covers the entire real line.

4.10 Uniform estimates over entropy time

At a time knot,

$$U(t_m, 0) = \frac{C_{m,-1} + C_{m,0} + 2S_0 E(t_m)}{2S_0}.$$

On interval m , Eq. (63) gives

$$U(t, 0) = (1 - \tau)U(t_m, 0) + \tau U(t_{m+1}, 0) + \frac{\delta_m^- - \delta_m^+}{2S_0} \tau (1 - \tau).$$

Let $[l, r]$ be the intersection of that interval with the desired objective range. The excess above the chord joining $U(l, 0)$ and $U(r, 0)$ is at most

$$\epsilon_m := \frac{\max\{\delta_m^- - \delta_m^+, 0\}}{8S_0} \frac{(r - l)^2}{(t_{m+1} - t_m)^2}. \quad (70)$$

Consequently it suffices to check the two rational inequalities

$$(U(l, 0) + \epsilon_m)^2 \leq c^2 l, \quad (U(r, 0) + \epsilon_m)^2 \leq c^2 r, \quad (71)$$

with nonnegative augmented values. Indeed, the affine interpolant of these augmented values majorizes $U(t, 0)$, and the affine interpolant of $c\sqrt{l}, c\sqrt{r}$ is at most $c\sqrt{t}$. This proves the objective over the entire interval, including a partial first or last interval, not only at the stored time knots.

Lemma 4.22 (Exact certified potentials). *The coefficient recurrences in Appendix A, the scalar inputs in Eq. (93), and the integer offset construction define global potentials satisfying Lemma 4.21. Their full time domains are $[0, 3/5]$ and $[0, 20]$. They satisfy*

$$U^{(1)}(t, 0) < c_j \sqrt{t} \quad (13/100 \leq t \leq 3/5), \quad (72)$$

$$U^{(2)}(t, 0) < \frac{361}{1000} \sqrt{t} \quad (3/5 \leq t \leq 20). \quad (73)$$

Proof. Generate the two coefficient arrays and all scalar inputs by the recurrences in Appendix A. Apply the deterministic offset algorithm above, including both time endpoints and the exact affine tails. The finite calculation gives

	N	Time knots	Upper bound on the augmented ratio
Fine	1000	2242	0.363270160722
Coarse	200	1418	0.359472078269

The first ratio is less than c_j and the second is less than $361/1000$. These are outward bounds on the maximum in Eq. (71), over all intervals meeting the respective objective ranges. The preliminary tests or the exact polynomial tests certify every real state and admissible jump; Eq. (64) extends the endpoint inequalities to every intermediate time. Finally, Eq. (70) includes the full quadratic bulge in the objective checks. All input constants, rounding rules, and finite verification formulas appear in the paper. $\square$

We apply Lemma 4.18 to treat all $T \geq 20$.

For one normalized stage, the bracket is at most $(\alpha\beta + \alpha^2/2)/K^2 \leq (\kappa_a + 1/2)\alpha^2/K^2$, obtained by integrating $\lambda(z)h(z)^2$ before its jump. Summing the stage bounds and using $\sum_i \alpha_i^2 \leq \rho_a \ell$ gives, for $L := S/K$,

$$\text{jump size} \leq \log 2, \quad \langle L \rangle_n \leq \frac{\kappa_a + 1/2}{K} \rho_a T \leq (\log 2)T.$$

Apply Eq. (54) with $V := (\log 2)T$, $b_1 := \log 2$, and $\delta := V/16$. For all $T \geq 20$,

$$\frac{\mathbb{E}_Q[(S/K)_+]}{\sqrt{T}} \leq \varphi(0)\sqrt{17 \log 2/16} + \frac{\varphi(1) \log 2 \log 17}{6\sqrt{T}} < \frac{361}{1000} < c_j. \quad (74)$$

The following finite rational calculation proves this:

$$\varphi(0) < 399/1000, \quad \varphi(1) < 121/500, \quad \log 2 < 347/500, \quad \log 17 < 1417/500.$$

For $0 < z < 1$, put $F(z) := \sum_{j=0}^{29} (-1)^j z^{2j+1}/(2j+1)$. The alternating-series bounds give $F(z) \leq \arctan z \leq F(z) + z^{61}/61$. Hence

$$\begin{aligned} \pi_- &:= 16F(1/5) - 4[F(1/239) + 239^{-61}/61] \leq \pi, \\ e_- &:= \sum_{j=0}^{15} \frac{1}{j!} < e. \end{aligned}$$

Indeed, Machin's identity is $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$: if $\theta := \arctan(1/5)$, then $\tan(2\theta) = 5/12$ and $\tan(4\theta) = 120/119$, and the subtraction formula gives $\tan(4\theta - \arctan(1/239)) = 1$ with the angle in $(0, \pi/2)$. The density bounds follow from the exact rational comparisons

$$2\pi_-(399/1000)^2 > 1, \quad 2\pi_-e_-(121/500)^2 > 1.$$

The logarithmic enclosures in Appendix A.1 give $\ell_+(2) < 347/500$ and $\ell_+(17) < 1417/500$. Finally define

$$R(x) := \frac{1 + \lfloor \sqrt{[10^{80}x]} \rfloor}{10^{40}} \quad (x \geq 0).$$

Integer square roots give $R(x) \geq \sqrt{x}$, and the exact rational test is

$$\frac{399}{1000} R\left(\frac{17}{16} \frac{347}{500}\right) + \frac{121}{500} \frac{347}{500} \frac{1417}{500} \frac{R(1/20)}{6} < \frac{361}{1000}.$$

This proves the bound at $T = 20$; the displayed expression decreases thereafter.

Proof of Theorem 4.19. Fix arbitrary a, n, u, f in the stated ranges. If $0 < T \leq 13/100$, Markov's inequality gives

$$y\sqrt{\ell} \leq \sqrt{\ell} \leq \sqrt{13/100}\sqrt{K} < c_j\sqrt{K}.$$

If $13/100 \leq T \leq 3/5$, apply the fine potential in Lemma 4.22. If $3/5 \leq T \leq 20$, apply the coarse potential. If $T \geq 20$, use Eq. (74). In each of these three ranges,

$$\mathbb{E}_Q[S_+] \leq c_j K \sqrt{T} = c_j \sqrt{K\ell}.$$

Substituting in Eq. (58) proves Eq. (56). All adjacent regimes are closed at their common boundaries. The small-jump proof includes arbitrarily small jumps, the exact tails cover all spatial states,

and the quadratic-time check covers all intermediate times. Empty tails give $y = 0$; zero stages were included explicitly. Thus no boundary regime is omitted.

Since $y \leq 1$, taking the supremum over all admissible n, u, f gives the first bound in Eq. (57). Divide it by $\sqrt{\kappa_a}$ and let $a \uparrow 1$. The factors $\sqrt{(\kappa_a + 1)/\kappa_a}$ and $1/\sqrt{\kappa_a}$ tend to one and zero, respectively. This proves the limsup bound without interchanging a supremum with any unproved limit. Inputs, dimensions, and thresholds may depend arbitrarily on a . $\square$

The earlier entropy-clock coefficient 9247/25000 follows immediately by replacing c_j with this larger number in Eq. (56) and Eq. (57).

Corollary 4.23 (A threshold-dependent bound). *Let U_a be defined in Eq. (3). For $\ell > 0$ define*

$$\Theta_a(\ell) := \frac{(1 + \sqrt{1 + 4c_j \sqrt{(\kappa_a + 1)\ell}})^2}{4\ell}. \quad (75)$$

Then, with $\ell := \log u$,

$$\psi_{a,n}(u) \leq \min\{1, \mathcal{R}(H(a)/\ell), \\ U_a/\sqrt{\ell}, \Theta_a(\ell)\}.$$

Also $C_*(a) \leq U_a$. The first branch of U_a can use $H(a)$ instead of J_a .

Proof. Propositions 4.6, 4.8, and 4.17, together with $H(a) \leq J_a$ and $\Gamma_B < \hat{\gamma}$, give the first three branches of U_a and the bound $\mathcal{R}(H(a)/\ell)$. The fourth branch is Theorem 4.19; the final branch follows from Theorem 4.24, proved below. Setting $z := \sqrt{y}$ in Eq. (56) and solving $\sqrt{\ell}z^2 - z - c_j\sqrt{\kappa_a + 1} \leq 0$ gives $y \leq \Theta_a(\ell)$. Taking the appropriate suprema proves the statements. For an input of positive mean, apply them to $f/\mathbb{E}_\mu[f]$. $\square$

The exact coefficient in Eq. (55) is a proved uniform upper coefficient; its optimality as a value of the control problem is not asserted.

4.11 The aligned score and clock concentration

The latent coordinate separates the entropy score into two martingales with different conditional variances. We charge the larger-variance part a stronger entropy cost and control the resulting random clock. Throughout this subsection and Section 4.12, put

$$\kappa := \kappa_a, \quad K := \kappa + 1, \quad q := 1 + 1/\kappa, \quad \gamma_a := \rho_a q^2, \quad \theta_a := \rho_a/q.$$

For $\epsilon > 0$ define

$$B_a(\epsilon) := c_A \sqrt{K(q + \epsilon)} + \sqrt{\frac{2\theta_a \gamma_a}{e\epsilon}} \exp\left(-\frac{K\epsilon^2}{32\gamma_a}\right) + \frac{1}{2} \sqrt{\frac{\rho_a}{\kappa}}. \quad (76)$$

Here e in an exponential bound denotes Euler's number.

Theorem 4.24. *For every $n \geq 1$, $0 < a < 1$, $u > 1$, and $f \geq 0$ with $\mathbb{E}_\mu[f] = 1$, put $\ell := \log u$ and $y := u \Pr_\mu[T_a f \geq u]$. Then, for every $\epsilon > 0$,*

$$y\sqrt{\ell} \leq B_a(\epsilon) + \sqrt{y}. \quad (77)$$

Consequently,

$$C_*(a) \leq 1 + \inf_{\epsilon > 0} B_a(\epsilon), \quad \limsup_{a \uparrow 1} \frac{C_*(a)}{\sqrt{\kappa_a}} \leq c_A. \quad (78)$$

The previously proved finite-noise bounds remain available by taking their minimum with this new bound.

Use the joint law, predictable coefficients, and double barrier from Lemma 4.9, so $0 \leq s_i \leq a$. Write $e_i := e(s_i, r_i)$, $\alpha_i := e_i/s_i$ for $s_i > 0$, and $\alpha_i := 0$ otherwise. Those arguments give

$$y\ell \leq \mathbb{E}_Q[S_+] + \sqrt{y\ell}, \quad \sum_i e_i \leq \ell, \quad \sum_i \alpha_i^2 \leq \rho_a \ell, \quad 0 \leq \alpha_i \leq d(s_i)/s_i \leq \log 2, \quad (79)$$

with the continuous convention at zero. Put $\sigma_i := \text{sgn}(b_i)$, using $\sigma_i := 1$ when $b_i = 0$, and let

$$I_i := \mathbf{1}_{\{\sigma_i Z_i = 1\}}, \quad S^H := \sum_i I_i S_i, \quad S^L := \sum_i (1 - I_i) S_i.$$

In the filtration revealing Z initially, I_i is predictable and both sums are martingales. Conditionally on Z and the observed prefix, S_i takes values $\alpha_i, -\beta_i$, with respective probabilities $\beta_i/(\alpha_i + \beta_i)$ and $\alpha_i/(\alpha_i + \beta_i)$, where $\beta_i := \kappa \alpha_i$ if $I_i = 1$ and $\beta_i := \alpha_i/\kappa$ otherwise. Zero coefficients give zero increments. Orthogonality and centering yield

$$\mathbb{E}_Q[(S^L)^2] = \frac{1}{\kappa} \mathbb{E}_Q\left[\sum_i (1 - I_i) \alpha_i^2\right] \leq \frac{\rho_a \ell}{\kappa}, \quad \mathbb{E}_Q[(S^L)_+] \leq \frac{1}{2} \sqrt{\frac{\rho_a \ell}{\kappa}}. \quad (80)$$

The last factor uses $\mathbb{E}[V_+] = \mathbb{E}[|V|]/2$ for centered V .

Embed the aligned part by the one-jump construction of Lemma 4.11, and divide the score by K to obtain a martingale L , starting from $L_0 := 0$. On an aligned stage with $\alpha_i > 0$, the unnormalized stage score rises at speed α_i before a jump to $-\beta_i$, with intensity $\lambda_i(z) := \alpha_i/(\beta_i + \alpha_i z)$ for elapsed time $0 \leq z \leq 1$. Its survival function is $\beta_i/(\beta_i + \alpha_i z)$, giving exactly the conditional endpoint law above. Use fresh randomness at each stage and reveal X_i at its endpoint. On unaligned stages, or stages with $\alpha_i = 0$, L is fixed, but X_i is still sampled with its original conditional law. This realizes the full adaptive joint law even when the latent coordinates are correlated.

Start a continuous clock at $\mathcal{A}_0 := 0$. On aligned stages let it run throughout the stage, including after its jump, at rate $2e_i/[K(1 + s_i)]$. On the other stages use rate zero. Thus

$$\mathcal{A}_n = H/K, \quad H := \sum_i \frac{2e_i I_i}{1 + s_i}.$$

Before an aligned jump its size and the clock rate satisfy

$$h = \frac{\alpha_i(\kappa + z)}{K} \leq \alpha_i \leq \frac{d(s_i)}{s_i}, \quad \dot{\mathcal{A}} = \lambda_i(z) \frac{2s_i}{1 + s_i} h.$$

The positive series for $d(s)/s^2$ implies

$$\frac{h(1 + s_i)}{2s_i} \leq \frac{d(s_i)(1 + s_i)}{2s_i^2} \leq \frac{(1 + a)\rho_a}{2} = \theta_a \leq \log 2.$$

After a jump the bracket rate is zero. Therefore, as measures on every path,

$$d\langle L \rangle \leq \theta_a d\mathcal{A}, \quad \text{every jump of } L \text{ has magnitude at most } \log 2. \quad (81)$$

We next control fluctuations of the aligned entropy clock.

Lemma 4.25. *There is a centered random variable W such that*

$$H \leq q\ell + W, \quad \mathbb{E}_Q[e^{tW}] \leq e^{\gamma_a \ell t^2/2} \quad (t \in \mathbb{R}). \quad (82)$$

Proof. Put $c_i := e_i/(1 + s_i)$, $Y_i := \sigma_i X_i$, and $\zeta_i := \sigma_i Z_i$. The function $d(s)/(1 + s)^2$ is increasing: its derivative has the sign of

$$(1 + s)d'(s) - 2d(s) = (1 - s) \operatorname{arctanh} s - \log(1 - s^2) \geq 0.$$

Since $e_i \leq d(s_i)$, pathwise

$$\sum_i c_i^2 \leq V_c, \quad V_c := \frac{d(a)}{(1 + a)^2} \ell.$$

Define $A_0 := \sum_i c_i(Y_i - s_i)$ and $B_0 := \sum_i c_i(Y_i - a\zeta_i)$. The first sum is a martingale sum in the observed-prefix filtration, and the second in the filtration revealing all Z initially. Each centered increment is supported in an interval of length $2c_i$. Its conditional moment generating function is at most $\exp(t^2 c_i^2/2)$: the second derivative of its log moment generating function is a tilted variance, at most c_i^2 , and integration from zero proves the bound. Iterating the exponential supermartingale and using the deterministic bound V_c gives

$$\mathbb{E}_Q[e^{tA_0}] \leq e^{t^2 V_c/2}, \quad \mathbb{E}_Q[e^{tB_0}] \leq e^{t^2 V_c/2}.$$

Both variables are centered. For $W := (A_0 - B_0)/a$, Cauchy–Schwarz gives

$$\mathbb{E}_Q[e^{tW}] \leq \mathbb{E}_Q[e^{2tA_0/a}]^{1/2} \mathbb{E}_Q[e^{-2tB_0/a}]^{1/2} \leq e^{2t^2 V_c/a^2} = e^{\gamma_a \ell t^2/2}.$$

No independence or common martingale-difference filtration for A_0, B_0 is required. Finally, with $D := \sum_i e_i$,

$$H - D = \sum_i c_i(\zeta_i - s_i) = W + \frac{1 - a}{a} \sum_i c_i s_i.$$

Since $s_i/(1 + s_i) \leq a/(1 + a)$, this is at most $W + D/\kappa$. Thus $H \leq qD + W \leq q\ell + W$. $\square$

Write $T := \ell/K$. For $\delta > 0$ and $r := qT + \delta$, Eq. (82) also gives

$$\mathbb{E}_Q[(\mathcal{A}_n - r)_+] \leq \frac{\gamma_a T}{eK\delta} \exp(-\frac{K\delta^2}{2\gamma_a T}). \quad (83)$$

Indeed $x_+ \leq e^{tx}/(et)$ for $t > 0$. Apply this to $W - K\delta$ with $t := K\delta/(\gamma_a \ell)$ and divide by K .

4.12 Aligned-clock comparison and proof of the aligned bound

We first adapt the inverse-entropy estimate to the stronger cost of the aligned clock. The resulting inequality supplies the coefficients for the potential comparison below.

Lemma 4.26. *Suppose $v > 0$, $0 < s_0 \leq 1$, and $p > 0$ satisfy $d(s_0) \leq v s_0$ and $2ps_0 \geq v(1 + s_0)$. For every $0 < s \leq 1$ and $0 \leq h \leq \min\{v, d(s)/s\}$,*

$$h^2 \leq p \frac{2s}{1 + s} h.$$

Proof. For $s \geq s_0$, $h(1 + s)/(2s) \leq v(1 + s_0)/(2s_0) \leq p$. For $s < s_0$, use monotonicity of $d(s)(1 + s)/s^2$ to obtain the same bound. Multiplication by $2sh/(1 + s)$ includes $h = 0$. $\square$

Lemma 4.27. *Let $g(s) = d(s)/s$ for $0 < s \leq 1$, and let σ be its inverse on $(0, \log 2)$. Define*

$$\chi(h) = \begin{cases} 1/4, & h = 0, \\ h(1 + \sigma(h))/(2\sigma(h)), & 0 < h < \log 2, \\ h, & h \geq \log 2. \end{cases}$$

Then χ is increasing and convex. If $0 < s \leq 1$ and $0 \leq h \leq d(s)/s$, then

$$h^2 \leq \chi(h) \frac{2s}{1+s} h.$$

If $v > 0$, $0 < s_0 \leq 1$, $d(s_0) \leq vs_0$, and $2ps_0 \geq v(1 + s_0)$, then $\chi(v) \leq p$. Consequently, affine interpolation between such endpoint bounds majorizes χ on each interval.

Proof. Put $w(s) = -\log(1 - s^2)/2$. Then $g'(s) = w(s)/s^2 > 0$. At $h = g(s)$, differentiation gives

$$\chi'(h) = \frac{1}{2} + \frac{w(s) - d(s)}{2sw(s)}, \quad \chi''(h) = \frac{s^2 d(s)/(1 - s^2) - 2w(s)^2}{2w(s)^3} \geq 0.$$

Indeed $d(s) \geq s^2/2$ follows from its positive power series. The inequality $2 \log t \leq t - t^{-1}$ for $t \geq 1$, obtained by differentiation, gives $(1 - s^2)w(s)^2 \leq s^4/4$ upon setting $t = (1 - s^2)^{-1/2}$. The endpoint values and slopes are $1/4, 1/2$ at zero and $\log 2, 1$ at $\log 2$, respectively. Thus the stated extension is increasing and convex. If $h \leq g(s)$, then $\sigma(h) \leq s$, proving the displayed inequality; the endpoints follow by continuity. Similarly, $g(s_0) \leq v$ implies $s_0 \leq \sigma(v)$ when $v < \log 2$, and hence $\chi(v) \leq v(1 + s_0)/(2s_0) \leq p$. For $v \geq \log 2$ the same bound follows from $p \geq v$. Convexity places χ below every chord joining upper endpoint bounds. $\square$

Use a positive cost $p(h)$ satisfying

$$h^2 \leq p(h) \frac{2s}{1+s} h \quad (0 < s \leq 1, 0 \leq h \leq d(s)/s).$$

The coarse piecewise constant costs from Lemma 4.26 and the fine affine costs from Lemma 4.27 both have this property. Suppose U has the regularity of Lemma 4.21, satisfies $U(0, x) \geq x_+$ and $U_t \geq 0$, and

$$p_k[U(t, x - h) - U(t, x) + hU_x(t, x)] \leq h^2 U_t(t, x) \quad (84)$$

for every allowed jump. Then, for $\tau := \inf\{t : \mathcal{A}_t \geq r\} \wedge n$,

$$\mathbb{E}_Q[(L_\tau)_+] \leq U(r, 0). \quad (85)$$

To prove this, apply the chain rule and jump compensator to $U(r - \mathcal{A}_{t \wedge \tau}, L_{t \wedge \tau})$. The cost inequality makes its drift nonpositive before a jump; a nonpositive jump remainder is covered directly by $U_t \geq 0$. After a jump the score is fixed and the continuing clock contributes $-\dot{\mathcal{A}}U_t \leq 0$. On other stages both are constant. There are finitely many stages and jumps, bounded intensities and states for each fixed input, and a continuous clock. The time knots contribute no extra term; if the clock remains at a knot then the score is constant. Expectations and stopping are therefore justified exactly as in Lemma 4.21. Time monotonicity gives $U \geq x_+$, proving Eq. (85).

Lemma 4.28. *The finite constructions in Appendix B give potentials satisfying Eq. (84) and the preceding initial and regularity conditions. They obey*

$$\begin{aligned} U^f(t, 0) &< c_A \sqrt{t} & (1/8 \leq t \leq 499/1000), \\ U^c(t, 0) &< (89/250) \sqrt{t} & (499/1000 \leq t \leq 30). \end{aligned}$$

Their full time domains contain $[0, 499/1000]$ and $[0, 30]$, respectively.

Proof. The fine potential passes the cubic spatial and quadratic time tests in Appendix B.5. Its cubic objective correction is checked on every time interval in Appendix B.9. The coarse potential retains the quadratic tests in Sections 4.9 and 4.10. Appendix B supplies both finite constructions, their initial majorants, and the rational objective bounds. These tests cover every state, jump size, and intermediate time. $\square$

For $r \geq 30$, Eq. (81) and the Gaussian comparison in Lemma 4.18 instead give

$$\frac{\mathbb{E}_Q[(L_\tau)_+]}{\sqrt{r}} \leq \varphi(0)\sqrt{33 \log 2/32} + \frac{\varphi(1) \log 2 \log 33}{6\sqrt{30}} < \frac{89}{250} < c_A.$$

Here the bracket budget is $V := (\log 2)r$ and the extra Gaussian variance is $V/32$. To verify the strict scalar inequality, use the density bounds $\varphi(0) < 399/1000$ and $\varphi(1) < 121/500$ proved above, the printed logarithmic enclosure ℓ_+ and square-root enclosure R , and the rational comparison

$$\frac{399}{1000}R\left(\frac{33}{32}\ell_+(2)\right) + \frac{121}{500}\ell_+(2)\ell_+(33)\frac{R(1/30)}{6} < \frac{89}{250}.$$

Together with Lemma 4.28, this proves $\mathbb{E}_Q[(L_\tau)_+] \leq c_A\sqrt{r}$ for every $r \geq 1/8$.

Proof of Theorem 4.24. First suppose $T := \ell/K \geq 1/8$, choose $\delta > 0$, and stop at clock $r := qT + \delta$. The residual $V := L_n - L_\tau$ is centered. Orthogonality of stopped martingale increments and Eq. (81) give

$$\mathbb{E}_Q[V^2] = \mathbb{E}_Q[\langle L \rangle_n - \langle L \rangle_\tau] \leq \theta_a \mathbb{E}_Q[(\mathcal{A}_n - r)_+].$$

If the clock never reaches r , the residual vanishes; otherwise continuity gives $\mathcal{A}_\tau = r$. Centering, Cauchy–Schwarz, and Eq. (83) imply

$$\mathbb{E}_Q[V_+] \leq \frac{1}{2} \sqrt{\frac{\theta_a \gamma_a T}{eK\delta}} \exp\left(-\frac{K\delta^2}{4\gamma_a T}\right).$$

Since $(x+z)_+ \leq x_+ + z_+$, multiply the stopped comparison and this residual bound by $K/\sqrt{\ell}$ to obtain

$$\frac{\mathbb{E}_Q[(S^H)_+]}{\sqrt{\ell}} \leq c_A \sqrt{K} \sqrt{q + \delta/T} + \frac{1}{2} \sqrt{\frac{\theta_a \gamma_a}{e\delta}} \exp\left(-\frac{K\delta^2}{4\gamma_a T}\right).$$

Set $\delta := \epsilon T$. The second term decreases with $T > 0$, so replace T there by $1/8$. Add Eq. (80) and use Eq. (79) to prove Eq. (77). For $0 < T \leq 1/8$, Markov's inequality gives $y\sqrt{\ell} \leq \sqrt{K/8} < c_A\sqrt{K} \leq B_a(\epsilon)$. This includes the common endpoint. Empty tails and zero coefficients cause no exception.

Since $y \leq 1$, taking the suprema over all dimensions, inputs, and thresholds proves the finite bound in Eq. (78). Choose $\epsilon_a := \sqrt{8\gamma_a \log K/K}$. Using $\sqrt{1+x} \leq 1 + x/2$ gives the fully explicit bound

$$\begin{aligned} C_*(a) &\leq c_A \sqrt{K} + 1 + \frac{c_A \sqrt{K}}{2\kappa} + c_A \sqrt{2\gamma_a \log K} \\ &\quad + \sqrt{\frac{\theta_a}{e}} \left(\frac{\gamma_a}{2 \log K}\right)^{1/4} + \frac{1}{2} \sqrt{\frac{\rho_a}{\kappa}}. \end{aligned} \tag{86}$$

As $a \uparrow 1$, $\gamma_a, \theta_a \rightarrow \log 2$ and $K/\kappa \rightarrow 1$. Dividing by $\sqrt{\kappa}$ proves the claimed limsup. All bounds precede the suprema, so the dimension, density, and threshold may depend arbitrarily on a . $\square$

The new upper proof separates the score according to the latent coordinate and controls its aligned entropy clock by exponential martingale bounds. It uses certified inverse-entropy intervals, joint state and jump polynomial tests, and separate time-endpoint derivative requirements. Its rational spline majorants are verified on the full state, jump-size, and entropy-time domain. Optimality of the entropy-clock comparison and of the global convolution constant remains open.

5 Lower bound

Section 5.1 evaluates point-mass inputs, derives explicit lower bounds, and identifies their exact leading constant. Section 5.2 proves a uniform estimate for the largest binomial atom. Section 5.3 uses an antichain argument to bound the concentration of sums of independent variables by this binomial estimate. Section 5.4 proves the exact asymptotics for fixed-block tensor families and establishes the optimality of the point-mass leading constant within this class.

5.1 Point-mass constructions and explicit lower bounds

For integers $n \geq 1$ and $0 \leq r \leq n$, set

$$\ell_{n,r} := (n - r) \log(1 + a) + r \log(1 - a).$$

Define the point-mass lower profile by

$$\mathcal{L}(a) := \sup_{\substack{n \geq 1, 0 \leq r \leq n \\ \ell_{n,r} > 0}} \sqrt{\ell_{n,r}} e^{\ell_{n,r}} 2^{-n} \sum_{j=0}^r \binom{n}{j}. \quad (87)$$

Also put $p := (1 + a)/2$, $n_* := -1/(2 \log p)$, and

$$\mathcal{N}(a) := \{\max\{1, \lfloor n_* \rfloor\}, \max\{1, \lceil n_* \rceil\}\}.$$

The following theorem records two explicit lower bounds furnished by point masses and determines the exact leading constant of their optimized profile.

Theorem 5.1 (Formal version of Theorem 1.2). *For every $0 < a < 1$,*

$$\max\left\{\max_{n \in \mathcal{N}(a)} \sqrt{n \log(1 + a)} p^n, \frac{\sqrt{\kappa_a d(a)}}{a \sqrt{2\pi}}\right\} \leq \mathcal{L}(a) \leq C_*(a).$$

Furthermore,

$$\lim_{a \uparrow 1} \frac{\mathcal{L}(a)}{\sqrt{\kappa_a}} = \sqrt{\frac{\log 2}{2e}}. \quad (88)$$

Consequently,

$$\liminf_{a \uparrow 1} \frac{C_*(a)}{\sqrt{\kappa_a}} \geq \sqrt{\frac{\log 2}{2e}}.$$

Proof. Take $f_n(x) := 2^n \mathbf{1}_{\{x=(1,\dots,1)\}}$, so $\mathbb{E}_\mu[f_n] = 1$ and $T_a f_n(x) = \prod_{i=1}^n (1 + ax_i)$. At a vertex with r negative coordinates, this equals $e^{\ell_{n,r}}$. The non-strict tail at that value is exactly the Hamming ball with at most r negative coordinates. Its contribution to $C_*(a)$ is the expression in Eq. (87), proving $\mathcal{L}(a) \leq C_*(a)$. For a fixed nonempty tail, $u \sqrt{\log u}$ is increasing for $u > 1$, so optimizing over arbitrary thresholds is equivalent to optimizing over these attained output levels.

For $r = 0$, the expression becomes $\sqrt{n \log(1+a)} p^n$. Its logarithmic derivative, with n regarded as positive and real, is $1/(2n) + \log p$. It increases until n_* and decreases thereafter, so its maximum over positive integers occurs in $\mathcal{N}(a)$.

For the other explicit lower bound, fix a and choose integers $k_n = np + O(1)$, with $r := n - k_n$. Then $\ell_{n,r} = nd(a) + O_a(1)$ and

$$e^{\ell_{n,r}} \Pr[\text{Bin}(n, 1/2) = k_n] = \Pr[\text{Bin}(n, p) = k_n] \sim \frac{1}{\sqrt{2\pi np(1-p)}}.$$

The equality follows by substituting the binomial masses; the asymptotic follows from Stirling's formula. The ratio of the fair-binomial upper tail to its first atom tends to p/a : each successive atom ratio tends to $(1-p)/p$, and these ratios are uniformly bounded by a constant less than one for all sufficiently large n . Dominated convergence therefore sums the geometric series. Multiplication by $\sqrt{\ell_{n,r}}$ gives $\sqrt{\kappa_a d(a)}/(a\sqrt{2\pi})$.

For Eq. (88), apply Theorem 5.4 to the one-coordinate block $g := 2\mathbf{1}_{\{1\}}$. Its tensor powers are exactly f_n , and the threshold optimization above gives $\mathcal{C}_g(a) = \mathcal{L}(a)$ for every a . Here $h_B = \log 2$ and $\beta_B = 1$, so Eq. (90) gives the claimed limit. The fixed-block theorem is proved independently in Section 5.4 using only the binomial-atom and antichain lemmas. The lower bound on the normalized limit inferior of $C_*(a)$ now follows from $C_*(a) \geq \mathcal{L}(a)$ and the displayed limit for $\mathcal{L}(a)$. $\square$

5.2 A uniform binomial-atom estimate

For an integer $m \geq 1$ and $0 < \theta < 1$, let

$$b_m(\theta) := \max_{0 \leq j \leq m} \binom{m}{j} \theta^j (1-\theta)^{m-j}.$$

We need an estimate that remains valid when the number of trials varies with the small binomial parameter. The next lemma identifies the sharp constant uniformly over this number.

Lemma 5.2 (Uniform small-parameter binomial envelope). *One has*

$$\limsup_{\theta \downarrow 0} \sup_{m \geq 1} \sqrt{m\theta} b_m(\theta) = \frac{1}{\sqrt{2e}}. \quad (89)$$

Proof. For the lower bound, choose $m := \lfloor (2\theta)^{-1} \rfloor$ for sufficiently small θ and use the atom at zero. Then $m\theta \rightarrow 1/2$ and $(1-\theta)^m \rightarrow e^{-1/2}$.

For the upper bound, it suffices to consider an arbitrary sequence $\theta \downarrow 0$ and arbitrary integers $m = m(\theta)$. Pass to a subsequence on which $\lambda := m\theta$ tends to zero, to some $\lambda_0 \in (0, \infty)$, or to infinity. If $\lambda \rightarrow 0$, use $b_m \leq 1$. If $\lambda \rightarrow \lambda_0 \in (0, \infty)$, the binomial modes are bounded, since a mode is $\lfloor (m+1)\theta \rfloor$ (with the usual possible adjacent tie). Their masses converge to Poisson masses. For a fixed integer $j \geq 0$,

$$\sup_{\lambda > 0} \sqrt{\lambda} e^{-\lambda} \frac{\lambda^j}{j!} = B_j := \frac{e^{-(j+1/2)} (j+1/2)^{j+1/2}}{j!}.$$

The maximum is attained at $\lambda = j + 1/2$. Writing $x := j + 1$, one gets

$$\log \frac{B_{j+1}}{B_j} = \int_{x-1/2}^{x+1/2} \log t dt - \log x < 0$$

by strict concavity of the logarithm. Thus $B_j \leq B_0 = 1/\sqrt{2e}$.

If $\lambda \rightarrow \infty$, both parameters at a binomial mode diverge. Stirling's formula, with its remainder applied to $m!$, $j!$, and $(m-j)!$, gives

$$b_m(\theta) \sim \frac{1}{\sqrt{2\pi m\theta(1-\theta)}}.$$

For precision, the mode differs from $m\theta$ by at most one; the associated relative-entropy exponent is $O(1/(m\theta(1-\theta))) = o(1)$, and each factorial remainder tends to zero. Hence this case contributes $1/\sqrt{2\pi} < 1/\sqrt{2e}$. Every subsequential possibility has been covered. If the supremum in Eq. (89) violated the claimed upper limit, selecting an almost-maximizing integer m for each member of a violating sequence would contradict this analysis. $\square$

5.3 Antichain concentration

The binomial-atom estimate controls concentration of sums through the following antichain argument.

Lemma 5.3 (An antichain concentration bound). *Let $V_1, \dots, V_m$ be independent copies of a real random variable whose law is a mixture of an upper component and a lower component, of probabilities $1-\theta$ and θ , respectively. Suppose every possible value of the upper component exceeds every possible value of the lower component by at least $h > 0$. Then, for every real t ,*

$$\Pr\left[\sum_{i=1}^m V_i \in [t, t+h]\right] \leq b_m(\theta).$$

Proof. Independently sample an upper value U_i , a lower value L_i , and a Bernoulli variable E_i of parameter θ for each coordinate, and set $V_i := U_i - E_i(U_i - L_i)$. Condition on all U_i, L_i . Changing a set of coordinates from upper to lower decreases the sum by at least h if the set is enlarged by even one coordinate. Thus the subsets of $[m]$ that put the sum in $[t, t+h)$ form an antichain.

For an antichain $\mathcal{A}$, a uniformly random maximal chain of the Boolean lattice contains any particular j -element set with probability $1/\binom{m}{j}$ and contains at most one member of $\mathcal{A}$. Hence

$$\sum_{j=0}^m \frac{|\mathcal{A} \cap \binom{[m]}{j}|}{\binom{m}{j}} \leq 1.$$

Under independent Bernoulli variables of common parameter θ , its probability is at most

$$b_m(\theta) \sum_{j=0}^m \frac{|\mathcal{A} \cap \binom{[m]}{j}|}{\binom{m}{j}} \leq b_m(\theta).$$

Averaging over the conditioned values proves the claim. The half-open interval ensures the assertion remains valid when a gap is exactly h . $\square$

5.4 Exact asymptotics for fixed-block tensor families

The following extension describes a substantially larger class of possible lower-bound constructions. Its scope is explicitly family-specific. Fix a positive integer k and a density g on $\{-1, 1\}^k$ with $g \geq 0$ and $\mathbb{E}_{\mu_k}[g] = 1$. Define

$$\mathcal{C}_g(a) := \sup_{m \geq 1} \sup_{u > 1} \sqrt{\log u} \sup_{\mu_{km}} \Pr[T_a(g^{\otimes m}) \geq u].$$

The block g and its dimension k are fixed as $a \uparrow 1$; the tensor exponent m and threshold u are unrestricted and may depend on a .

For a nonempty proper subset $B \subset \{-1, 1\}^k$, write

$$\alpha := \Pr_{\mu_k}[B], \quad h_B := \log(1/\alpha), \quad \beta_B := \frac{1}{|B|} \sum_{x \in B} |\{i : x^{(i)} \notin B\}|.$$

Thus β_B is the mean number of outward edges at a uniform point of B .

Theorem 5.4 (Exact fixed-block tensor asymptotics). *If $g = \mathbf{1}_B/\alpha$ for a nonempty proper fixed set B , then*

$$\lim_{a \uparrow 1} \frac{\mathcal{C}_g(a)}{\sqrt{\kappa_a}} = \sqrt{\frac{h_B}{2e\beta_B}} \leq \sqrt{\frac{\log 2}{2e}}. \quad (90)$$

Equality in the last inequality holds if and only if B is a subcube. If g has at least two distinct positive values, then

$$\lim_{a \uparrow 1} \frac{\mathcal{C}_g(a)}{\sqrt{\kappa_a}} = 0. \quad (91)$$

For $g \equiv 1$, $\mathcal{C}_g(a) = 0$.

Proof of Theorem 5.4. First take $g := \mathbf{1}_B/\alpha$. Put $\epsilon := (1 - a)/2$, $p := 1 - \epsilon$, and $g_a := T_a g$. The operator factorizes, so $T_a(g^{\otimes m}) = g_a^{\otimes m}$. Under the probability measure $Q_a := g_a \mu_k$, set $\theta_a := \Pr_{Q_a}[B^c]$. For a sufficiently close to one, $p^k > 1/2$ and

$$\min_{x \in B} g_a(x) \geq \frac{p^k}{\alpha}, \quad \max_{x \notin B} g_a(x) \leq \frac{1 - p^k}{\alpha}.$$

Indeed, the no-flip event contributes p^k/α at each point of B , whereas reaching B from outside requires at least one flip. Thus the upper and lower components of $\log g_a(X)$ under Q_a , according to membership in B , have gap at least

$$h_a := \log \frac{p^k}{1 - p^k} \longrightarrow \infty.$$

Both component probabilities are strictly between zero and one. Also, expansion in the number of flipped coordinates gives

$$\theta_a = \beta_B \epsilon + O_B(\epsilon^2), \quad \kappa_a \theta_a \longrightarrow \beta_B. \quad (92)$$

The first equality is also seen by sampling a uniform $Z \in B$: the one-flip exits contribute the average outward degree, and two or more flips have probability $O_B(\epsilon^2)$.

Let $L := \log(g_a^{\otimes m}(X))$ under $Q_a^{\otimes m}$. Applying Lemma 5.3 to each half-open logarithmic slab gives, for $\ell := \log u > 0$,

$$\begin{aligned} u \Pr_{\mu_{km}}[g_a^{\otimes m} \geq u] &= \mathbb{E}_{Q_a^{\otimes m}}[e^{\ell-L} \mathbf{1}_{\{L \geq \ell\}}] \\ &\leq \sum_{j \geq 0} e^{-jh_a} \Pr_{Q_a^{\otimes m}}[L \in [\ell + jh_a, \ell + (j+1)h_a)] \\ &\leq \frac{b_m(\theta_a)}{1 - e^{-h_a}}. \end{aligned}$$

For a nonempty tail, $\ell \leq mh_B$, because $g_a \leq 1/\alpha$. Consequently

$$\frac{\mathcal{C}_g(a)}{\sqrt{\kappa_a}} \leq \frac{\sqrt{h_B}}{\sqrt{\kappa_a \theta_a}(1 - e^{-h_a})} \sup_{m \geq 1} \sqrt{m \theta_a} b_m(\theta_a).$$

Use Eq. (92) and Lemma 5.2 to obtain the upper limit in Eq. (90). This estimate holds uniformly over every tensor exponent and every threshold, not only over a selected plateau or a prescribed dimension scaling.

For the matching lower limit, let $b_B(x) := |\{i : x^{(i)} \notin B\}|$ for $x \in B$. Uniformly over this fixed set,

$$\log g_a(x) = h_B - \epsilon b_B(x) + O_B(\epsilon^2).$$

Choose $m := \lfloor (2\beta_B \epsilon)^{-1} \rfloor$ and $\delta_\epsilon := \epsilon^{1/4}$. Under uniform measure conditioned on B^m , the variables $b_B(X_i)$ are independent, bounded, and have mean β_B . Since $m\epsilon\beta_B \rightarrow 1/2$ and $\text{Var}(\epsilon \sum_i b_B(X_i)) = O_B(\epsilon)$, Chebyshev's inequality and $m\epsilon^2 = O_B(\epsilon)$ give

$$\Pr_{\mu_{km}} [\log g_a^{\otimes m} \geq mh_B - \frac{1}{2} - \delta_\epsilon \mid B^m] \rightarrow 1.$$

The displayed threshold is positive for sufficiently small ϵ . Taking its exponential as u yields

$$u \Pr_{\mu_{km}} [g_a^{\otimes m} \geq u] \geq e^{-1/2 - o(1)}, \quad \frac{\log u}{\kappa_a} \rightarrow \frac{h_B}{2\beta_B}.$$

This proves the matching lower limit.

We include the edge-isoperimetric step and its equality cases. If $E(B)$ is the number of internal unoriented edges and $M := |B|$, induction on the cube dimension proves

$$2E(B) \leq M \log_2 M,$$

with the convention $0 \log_2 0 := 0$ for empty sections. Split B into coordinate sections of sizes $M_0 \leq M_1$. The cross-edge count is at most M_0 . Apply induction in each section. The remaining inequality is the binary-entropy inequality $H_2(t) \geq 2t$ for $t := M_0/(M_0 + M_1) \in [0, 1/2]$, which follows from strict concavity and the chord between 0 and $1/2$. Equality requires either one empty section, or equally sized identical sections, with equality in the induction hypotheses. Inductively these are exactly subcubes. Therefore

$$\beta_B = k - \frac{2E(B)}{|B|} \geq k - \log_2 |B| = \frac{h_B}{\log 2},$$

with precisely the asserted equality cases.

Finally suppose g has at least two distinct positive values. Partition its values at a gap between two consecutive positive levels, for example by separating its largest value from all lower values. Since $T_a g \rightarrow g$ uniformly on the fixed finite cube, the logarithms under $g_a \mu_k$ retain a fixed positive gap for a sufficiently close to one. The two mixture probabilities are bounded away from zero and one. The same antichain argument applies with a fixed gap $h > 0$ and $\theta_a \in [\eta, 1 - \eta]$ for some $\eta > 0$. Stirling's formula at a binomial mode, with the finitely many small m absorbed into the constant, gives

$$\sup_{\eta \leq \theta \leq 1-\eta} \sup_{m \geq 1} \sqrt{m} b_m(\theta) < \infty.$$

Since $g_a \leq \max g$, the argument bounds $\mathcal{C}_g(a)$ by a constant depending only on g , uniformly for a close to one. Dividing by $\sqrt{\kappa_a} \rightarrow \infty$ proves Eq. (91). The remaining constant density $g \equiv 1$ has no tail at $u > 1$. $\square$

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A A finite construction of the spline coefficients

Appendix A.1 specifies the exact arithmetic and scalar inputs from the printed rational constants. Appendix A.2 defines the time knots and coefficient rows for the fine and coarse constructions by finite recurrences. Appendix A.3 gives the selection and elimination rules that produce the candidate splines without taking stored coefficient tables as inputs. Appendix A.4 certifies these candidates using the polynomial tests of Section 4.9 and exact objective checks, without assuming convergence of an iterative numerical method.

A.1 Arithmetic and scalar inputs

Put $S := 10^{12}$ and $q := 2^{52}$. For integers a and $b > 0$, define

$$\text{rn}(a, b) := \lfloor a/b + 1/2 \rfloor, \quad a \odot b := \text{rn}(ab, q), \quad a \oslash b := \text{rn}(aq, b).$$

Thus ties are rounded toward positive infinity. Every addition, subtraction, and multiplication without $\odot$ is exact integer arithmetic. The operation $\oslash$ is used only with positive second argument.

Here are explicit rational logarithmic enclosures. For $1 \leq x \leq 2$, set $z := (x - 1)/(x + 1)$ and

$$L(x) := 2 \sum_{j=0}^{35} \frac{z^{2j+1}}{2j+1}, \quad H(x) := L(x) + \frac{2z^{73}}{73(1-z^2)}.$$

For a positive rational y , repeatedly multiply by two while $y < 1$, and divide by two while $y > 2$, recording $y = 2^e x$ with $1 \leq x \leq 2$. Define

$$\ell_+(y) := H(x) + \begin{cases} eH(2), & e \geq 0, \\ eL(2), & e < 0. \end{cases}$$

The geometric-series remainder gives $\log y \leq \ell_+(y)$. Put $d_+(0) := 0$, $d_+(1) := H(2)$, and

$$d_+(s) := \frac{(1+s)\ell_+(1+s) + (1-s)\ell_+(1-s)}{2} \quad (0 < s < 1).$$

For each $N \in \{1000, 200\}$ and $0 \leq k \leq \lfloor 693147181N/10^9 \rfloor$, put

$$h_k := \min\{(k+1)/N, 693147181/10^9\}.$$

Starting with $(l, r) := (0, S+1)$, repeat the following step until $r-l=1$: set $b := \lfloor (l+r)/2 \rfloor$; replace l by b if $d_+(b/S) \leq h_k b/S$, and otherwise replace r by b . Denote the final l by b_k . Set

$$P_k := \lceil 10^8 S h_k / b_k \rceil. \quad (93)$$

This gives the scalar inputs in Eq. (59). The finite checks are $b_k > 0$, $d_+(b_k/S) \leq h_k b_k/S$, and $10^8/2 \leq P_k \leq 7 \cdot 10^8/10$.

A.2 Time knots and coefficient rows

Use the following two parameter choices. The variable η selects the fine construction when it is one and the coarse construction when it is zero; K is the largest control used to construct a row, independently of the jump bins used to verify it.

	η	N	K	$k_{\min}$	A_{end}	M
Fine	1	1000	693	1	$3S/5$	16
Coarse	0	200	139	16	$20S$	12

There are $n := 8000$ spatial entries, indexed by $i = 0, \dots, n-1$, with centers $x_i := (i-4000+1/2)/N$. Construct the integer time knots from $A_0 := 0$ by $A_{m+1} := A_m + a_m$, stopping at A_{end} , where

$$a_m := \begin{cases} \min\{A_{\text{end}} - A_m, 2 \cdot 10^9, \max\{10^4, \text{rn}(3(A_m + 10^6), 500)\}\}, & \eta = 1, \\ \min\{A_{\text{end}} - A_m, 2 \cdot 10^{10}, \max\{10^7, \text{rn}(3(A_m + 10^9), 200)\}\}, & \eta = 0. \end{cases}$$

The rates used in selecting a row are the integers

$$r_k := \begin{cases} \text{rn}(qSN, kb_{k-1}), & \eta = 1, \\ \text{rn}(q(100N^2 + 81k^2), 200k^2), & \eta = 0, \end{cases} \quad k_{\min} \leq k \leq K.$$

Initialize $w_i := \text{rn}(q \max\{2(i-4000)+1, 0\}, 2N)$ and $C_{0,i-4000} := S \max\{2(i-4000)+1, 0\}/(2N)$. The integers w_i are retained at scale q between time steps.

At each time step, save $w^{\text{old}} := w$ and apply the selection rule below to obtain (π, v) . Repeat the following procedure at most M times: replace w by the elimination rule below with inputs $(\pi, a_m, w^{\text{old}})$, and apply the selection rule to this new w , obtaining (π', v') . Stop the repetitions if

$$10^9 \max_i |w_i - w_i^{\text{old}} - \text{rn}(a_m v'_i, S)| \leq q. \quad (94)$$

Otherwise put $\pi := \pi'$ and continue, retaining the last w even if the test has not passed after M repetitions. Set

$$C_{m+1,i-4000} := \text{rn}(S w_i, q).$$

The cap on the number of repetitions makes this a finite definition. Eq. (94) is part of this definition and is not a hypothesis used to certify the resulting spline.

A.3 Selection and elimination

For the selection rule, initialize $(\pi_i, v_i) := (-1, 0)$ at every index. For each $K + 2 \leq i \leq n - 3$, set

$$z_i := \text{rn}(N^2(w_{i+1} - 2w_i + w_{i-1}), 4).$$

If $z_i \geq 0$, replace (π_i, v_i) by $(0, z_i)$. Next visit $k = k_{\min}, \dots, K$ in increasing order and compute

$$g_{ik} := \begin{cases} \text{rn}(r_k[2(w_{i-k} - w_i) + k(-3w_i + 4w_{i+1} - w_{i+2})], 2q), & \eta = 1, \\ \text{rn}(r_k[w_{i-k} - w_i + k(w_{i+1} - w_i)], q), & \eta = 0. \end{cases}$$

Replace (π_i, v_i) by (k, g_{ik}) precisely when $g_{ik} > v_i$. This strict comparison specifies all ties.

For the elimination rule, initialize three integer arrays $(B_i, H_i, V_i) := (0, 0, 0)$. Visit $i = 0, \dots, n - 1$ in increasing order. If $\pi_i = -1$, set $V_i := w_i^{\text{old}}$ and go to the next index. Otherwise set $k := \max\{\pi_i, 1\}$. When $\pi_i = 0$, put

$$\beta := \text{rn}(a_m N^2 q, 4S), \quad (D, F, G) := (q + 2\beta, -\beta, 0).$$

When $\pi_i > 0$, put $\beta := \text{rn}(a_m r_k, S)$ and

	α	(D, F, G)
$\eta = 1$	$\text{rn}(k\beta, 2)$	$(q + \beta + 3\alpha, -4\alpha, \alpha)$
$\eta = 0$	$k\beta$	$(q + \beta + \alpha, -\alpha, 0)$.

Initialize $(c, e, y) := (-\beta, 0, w_i^{\text{old}})$. For $j = i - k, \dots, i - 1$, simultaneously replace these three integers by

$$(c, e, y) \mapsto (e - c \odot B_j, -c \odot H_j, y - c \odot V_j).$$

Then put $d := D + c$ and set

$$B_i := (F + e) \odot d, \quad H_i := G \odot d, \quad V_i := y \odot d.$$

In these two finite constructions all such divisors satisfy $d > q/100$. Finally visit $i = n - 1, \dots, 0$ in decreasing order and set

$$w_i := V_i - B_i \odot w_{i+1} - H_i \odot w_{i+2},$$

where terms whose indices exceed $n - 1$ are omitted. This completes the definition of every stored coefficient.

A.4 Finite certification

Extend the rows by the exact affine tails specified in Eq. (62). Construct both endpoint offsets in increasing order of the spatial and jump indices, using the preliminary and polynomial tests in Section 4.9, and integrate them by Eq. (63). These are finite integer operations on the arrays defined above. Finally check Eq. (71) on every time interval meeting the indicated objective range.

The fine and coarse constructions have 2242 and 1418 time knots. The scalar bisections satisfy all tests after Eq. (93), and the row constructions finish with positive elimination divisors. Applying the offset algorithm and the rational objective tests gives the outward bounds in Lemma 4.22. These computations prove Eqs. (72) and (73). Thus the recurrences specify complete certificates from the printed constants alone; checking them requires a finite arithmetic calculation, without access to stored coefficient data.

B A finite construction for the aligned clock

We specify both potentials in Lemma 4.28 by finite integer recurrences. The coarse construction uses the quadratic tests in Section 4.9; the fine construction uses cubic spatial splines and quadratic interpolation in time. Here N and n specify the spline array, independently of the Boolean dimension. All scalar inputs, rows, and offsets are determined below. No stored coefficient array or solver convergence assertion is an input.

B.1 Scalar inputs and the coarse construction

Put $S := 10^{12}$ and $G := 10^8$. Use the logarithmic enclosure d_+ and arithmetic operations in Appendix A.1. For the coarse construction retain $q := 2^{52}$, set $N := 200$, and construct h_k, b_k by the bisections in that appendix. Replace Eq. (93) by

$$P_k = \lceil Gh_k(S + b_k)/(2b_k) \rceil, \quad p_k = P_k/G.$$

The checks $b_k > 0$, $d_+(b_k/S) \leq h_k b_k/S$, and $G/4 \leq P_k \leq 7G/10$ hold and imply Lemma 4.26 on each coarse jump bin.

The coarse rows have $n = 8000$ entries, first spatial index -4000 , and largest construction control $J = 138$. Start from the same $w_i, C_{0,i-4000}$ as in Appendix A.2, and set $A_0 = 0$, $A_{m+1} = A_m + a_m$, where

$$a_m = \min\{30S - A_m, 2 \cdot 10^{10}, \max\{10^4, \text{rn}(3(A_m + 25000000), 200)\}\}.$$

Stop at $A_m = 30S$. Use $k_{\min} = 1$, the rules marked $\eta = 1$ in Appendix A.3, and rates

$$r_k = \text{rn}(qN(S + b_{k-1}), 2kb_{k-1}), \quad 1 \leq k \leq J.$$

Replace K by J , the divisor 4 in the central selection value by 8, and the central elimination divisor $4S$ by $8S$. All other selection, elimination, and back-substitution rules remain the same. At every step use $M = 16$ and Eq. (94); set $C_{m+1,i-4000} := \text{rn}(Sw_i, q)$. This gives 2166 knots, with every elimination divisor greater than $q/100$. The coarse potential retains the quadratic spatial spline, affine time interpolation, exact affine tails, and zero-initialized offset construction in Section 4.9.

For the fine construction put $q := 2^{56}$, $N := 1000$, $J := 693$, and $\nu := 3/16$. Its cost endpoints are

$$v_j = j/N \quad (0 \leq j \leq J), \quad v_{J+1} = (J + \nu)/N.$$

For each $1 \leq j \leq J + 1$, perform the integer bisection from Appendix A.1 with h_k replaced by v_j , and subtract two from its final lower endpoint. Denote the resulting integer by b_j . Set

$$P_0 = G/4, \quad P_j = \lceil Gv_j(S + b_j)/(2b_j) \rceil \quad (1 \leq j \leq J + 1). \quad (95)$$

The finite checks are

$$0 < b_j \leq S, \quad d_+(b_j/S) \leq v_j b_j/S, \quad G/4 = P_0 \leq P_1 \leq \dots \leq P_{J+1} \leq 7G/10.$$

Let $\bar{p}$ interpolate the values P_j/G affinely between these endpoints. Lemma 4.27 gives $\bar{p}(h) \geq \chi(h)$ throughout $[0, v_{J+1}]$. The logarithmic enclosure also gives $\log 2 < 693147181/10^9 < v_{J+1}$.

B.2 The fine coefficient recurrence

There are initially $n = 6000$ entries, with first spatial index $i_0 = -3000$. Set

$$w_i = \text{rn}(q \max\{2(i + i_0) + 1, 0\}, 2N), \quad C_{0,i+i_0} = S \max\{2(i + i_0) + 1, 0\} / (2N).$$

Starting from $A_0 = 0$, set $A_{m+1} = A_m + a_m$ until $A_m = 18S/25$, where

$$a_m = \min\{18S/25 - A_m, c_m, \max\{1000, \text{rn}(A_m + 10^6, 250)\}\}, \quad c_m = \begin{cases} 250000000, & A_m < 23S/50, \\ 10^9, & A_m \geq 23S/50. \end{cases}$$

This gives 4618 knots. Put $r_k = \text{rn}(qN(S + b_k), 2kb_k)$ for $1 \leq k \leq J$. Apply the selection rule of Appendix A.3 with these rates, the three-point derivative, central divisor 8, and active indices $J + 3 \leq i \leq n - 3$. After the controls $1, \dots, J$, also test control $J + 1$ with value

$$\text{rn}(N(13w_{i-J} + 3w_{i-J-1} - 16w_i), 16J + 3) + \text{rn}(N(-3w_i + 4w_{i+1} - w_{i+2}), 2). \quad (96)$$

As before, replace the selected value only on strict improvement. All inactive indices have $(\pi_i, v_i) = (-1, 0)$.

At the start of a step select (π, v) from w and define the time-control row at scale q by

$$z_i = w_i + \text{rn}(a_m v_i, 2S).$$

Save $R_i^0 = \text{rn}(Sw_i, q)$ and $R_i^1 = \text{rn}(Sz_i, q)$. Apply at most 20 elimination-and-selection iterations with the fixed right-hand side z . For each elimination use the recurrence and back-substitution in Appendix A.3, with the following replacements. If $\pi_i = -1$, set $V_i = z_i$. For the other controls put

$$\beta := \begin{cases} \text{rn}(a_m N^2 q, 16S), & \pi_i = 0, \\ \text{rn}(a_m r_k, 2S), & \pi_i = k \in \{1, \dots, J\}, \\ \text{rn}(a_m Nq, 2S(16J + 3)), & \pi_i = J + 1. \end{cases}$$

For $1 \leq \pi_i = k \leq J$ set $\alpha := \text{rn}(k\beta, 2)$, and for $\pi_i = J + 1$ set $\alpha := \text{rn}(a_m Nq, 4S)$. Initialize

$$(D, F, G) := \begin{cases} (q + 2\beta, -\beta, 0), & \pi_i = 0, \\ (q + \beta + 3\alpha, -4\alpha, \alpha), & 1 \leq \pi_i \leq J, \\ (q + 16\beta + 3\alpha, -4\alpha, \alpha), & \pi_i = J + 1. \end{cases}$$

For controls $0, 1, \dots, J$, initialize $(c, e, y) = (-\beta, 0, z_i)$ and begin elimination at $i - \max\{\pi_i, 1\}$. For control $J + 1$, initialize $(c, e, y) = (-3\beta, -13\beta, z_i)$ and begin at $i - J - 1$. Finish the same forward elimination and back-substitution. Every divisor is greater than $q/100$.

Select (π', v') from the resulting w and stop the iterations if

$$10^{13} \max_i |w_i - z_i - \text{rn}(a_m v'_i, 2S)| \leq q.$$

Otherwise set $\pi = \pi'$ and repeat, retaining the last w after the cap of 20 iterations. Set $R_i^2 = \text{rn}(Sw_i, q)$, and retain w for the next step. Thus R^2 of one interval is R^0 of the next. Output entry i has spatial index $i + i_0$. The stopping test only defines the rows; the independent polynomial inequalities below certify them.

B.3 A finite positive right-tail extension

Apply the following rule separately to every fine endpoint and time-control row. Write its indices in spatial coordinates and put

$$j_0 = 2799, \quad M = 2000, \quad F_i := S \max\{2i + 1, 0\}/(2N).$$

Define

$$a := \max\{0, C_{j_0} - F_{j_0}\}, \quad b = \min\{-1, C_{j_0} - C_{j_0-1} - S/N\}.$$

Keep C_i for $i \leq j_0$. Set

$$\beta = \begin{cases} \text{rn}(M^4 \text{rn}(qa, a - b), (M - 1)^4), & a > 0, \\ 0, & a = 0, \end{cases} \quad z_0 = q, \quad z_r = z_{r-1} \odot \beta.$$

For $1 \leq r < M$, replace the coefficient at $i = j_0 + r$ by

$$F_i + \text{rn}(a \text{rn}(z_r(M - r)^4, M^4), q). \quad (97)$$

Set every coefficient with $i \geq j_0 + M$ equal to F_i . Store the resulting 8000 entries with first index -3000 , and extend outside that interval by F_i . The rule gives an exactly affine right tail and a nonnegative excess on its modified part. Its effect on every derivative and jump is included in the tests below.

B.4 The cubic fine potential

Put $\Delta := 1/N$. An integer coefficient C_i represents the value C_i/S at center $(i + 1/2)\Delta$. On the spatial cell $x = (i + 1/2 + r)\Delta$, $0 \leq r \leq 1$, set

$$\begin{aligned} P(t, x) = & \frac{(1 - r)^3}{6} p_{i-1}(t) + \frac{3r^3 - 6r^2 + 4}{6} p_i(t) \\ & + \frac{-3r^3 + 3r^2 + 3r + 1}{6} p_{i+1}(t) + \frac{r^3}{6} p_{i+2}(t). \end{aligned} \quad (98)$$

All four weights are nonnegative, sum to one, and have weighted center x . Direct substitution at $r = 0, 1$ shows agreement of the value and the first two spatial derivatives across cells. In particular, P is C^2 in space. Its curvature is linear on each cell, with endpoint values

$$\frac{N^2}{S} (C_{i-1} - 2C_i + C_{i+1}), \quad \frac{N^2}{S} (C_i - 2C_{i+1} + C_{i+2}). \quad (99)$$

No convexity of coefficient rows is assumed.

Initial coefficients are exactly $C_i(0) = S((i + 1/2)\Delta)_+$. Jensen's inequality for the four weights proves $P(0, x) \geq x_+$. Outside the stored interval, every row is extended by these exact fixed affine tails: zero on the left and $(i + 1/2)\Delta$ on the right. This is a global definition, not an assumed decay property of a numerical approximation.

On time interval $[t_m, t_{m+1}]$, write

$$t_m = A_m/S, \quad a_m = A_{m+1} - A_m, \quad \tau = (t - t_m)/(t_{m+1} - t_m).$$

The finite recurrence above supplies endpoint rows R^0, R^2 and one temporal Bernstein control row R^1 . The control row is not an assertion about the function value at the midpoint. Define

$$p_i(t) = \frac{(1 - \tau)^2 R_i^0 + 2\tau(1 - \tau) R_i^1 + \tau^2 R_i^2}{S}. \quad (100)$$

Set

$$V^0 = 2(R^1 - R^0), \quad V^1 = R^2 - R^0, \quad V^2 = 2(R^2 - R^1).$$

The time derivative is a degree-two Bernstein combination of the spatial splines formed from V^e/a_m , for $e = 0, 1, 2$.

For each interval the finite rules below construct nonnegative integers $\delta_m^0, \delta_m^1, \delta_m^2$, whose sum is divisible by three. Starting from $E(0) = 0$, define

$$E(t) = E(t_m) + \frac{1}{S}[\delta_m^0 \tau + (\delta_m^1 - \delta_m^0) \tau^2 + (\delta_m^0 - 2\delta_m^1 + \delta_m^2) \tau^3 / 3], \quad (101)$$

so

$$E(t_{m+1}) - E(t_m) = \frac{\delta_m^0 + \delta_m^1 + \delta_m^2}{3S}.$$

The derivative E_t has degree-two Bernstein controls δ_m^e/a_m . Put $U = P + E$. The spatial jump remainder is independent of E ; it has temporal controls given by the remainders of R^e/S . Both sides of Eq. (84) are therefore degree-two Bernstein combinations. It suffices to prove the inequality separately for (R^e, V^e, δ_m^e) , for all three e .

B.5 Full state and jump polynomials

For an integer row C , put

$$A_i = C_{i-1} + 4C_i + C_{i+1}, \quad B_i = 3(C_{i+1} - C_{i-1}), \quad C_i^\circ = 3(C_{i-1} - 2C_i + C_{i+1}),$$

$$D_i = -C_{i-1} + 3C_i - 3C_{i+1} + C_{i+2}.$$

Then $6SP = A_i + B_i r + C_i^\circ r^2 + D_i r^3$ on cell i . In the coefficient formulas below abbreviate C_i° by C_i . A superscript V will indicate that the same cubic coefficients were formed from a temporal derivative-control row, not from the source row.

For a full jump bin $k \geq 1$, write $h = (k + v)\Delta$, $0 \leq v \leq 1$. Divide the square $0 \leq r, v \leq 1$ into the two closed triangles $v \leq r$ and $r \leq v$. On the first take $j = i - k, \epsilon = 0$; on the second take $j = i - k - 1, \epsilon = 1$. The exact raw remainder is

$$R_{i,k}(r, v) = A_j + B_j(r - v + \epsilon) + C_j(r - v + \epsilon)^2 + D_j(r - v + \epsilon)^3 - (A_i + B_i r + C_i r^2 + D_i r^3) + (k + v)(B_i + 2C_i r + 3D_i r^2). \quad (102)$$

It equals $6S[P(x - h) - P(x) + hP_x(x)]$. Write $V_i(r) = A_i^V + B_i^V r + C_i^V r^2 + D_i^V r^3$. For one temporal component, clearing denominators in Eq. (84) gives

$$SG(k + v)^2[V_i(r) + 6\delta] - a_m N^2[P_k + (P_{k+1} - P_k)v]R_{i,k}(r, v) \geq 0. \quad (103)$$

On either triangle substitute its barycentric coordinates and homogenize to degree five using $\lambda_0 + \lambda_1 + \lambda_2 = 1$. Nonnegativity of all 21 monomial coefficients proves the polynomial nonnegative on the entire triangle. Each coefficient multiplying δ is strictly positive, since $k \geq 1$. Thus each test supplies an explicit integer lower bound on δ , obtained by a ceiling after moving the other terms to the right. A negative numerator supplies no additional requirement.

For the final partial bin $0 \leq v \leq \nu$, replace the cost by $[P_J + (P_{J+1} - P_J)v/\nu]/G$. The relevant rectangle is covered by the triangles with vertices

$$(0, 0), (1, 0), (1, \nu); \quad (0, 0), (1, \nu), (\nu, \nu); \quad (0, 0), (0, \nu), (\nu, \nu).$$

The first two use $\epsilon = 0$; the last uses $\epsilon = 1$. The same degree-five coefficient test applies after clearing the positive denominators 16, 3. Equivalently, multiply by $16^5 \cdot 3$ and use integer numerator barycentric coordinates; the resulting coefficient tests involve only integers. This avoids imposing the unnecessarily larger domain $v \leq 1$ at the last bin.

B.6 Arbitrarily small jumps

When $h = v\Delta$, $0 \leq v \leq 1$, and $v \leq r$, the jump remains in one cell. Cancellation gives

$$\frac{P(x-h) - P(x) + hP_x(x)}{h^2} = \frac{N^2}{6S}(C_i + 3D_i r - D_i v).$$

The quotient has the displayed continuous extension at $h = 0$. Multiplication by the affine cost gives a polynomial of degree at most two; comparing it with the cubic $V_i(r) + 6\delta$ and homogenizing to degree three on $v \leq r$ gives ten nonnegative-coefficient tests.

When $r \leq v$, put $r = vz$, $z \in [0, 1]$. The C^2 matching identities give the exact cancellation

$$\frac{P(x-h) - P(x) + hP_x(x)}{h^2} = \frac{N^2}{6S}[C_i + v\{D_i(3z^2 - z^3) - D_{i-1}(1-z)^3\}]. \quad (104)$$

Consequently the necessary comparison is a polynomial of degree at most three in each of v, z . Checking its 16 tensor Bernstein coefficients is sufficient on the whole unit square. For transparency, index those coefficients by $j, l \in \{0, 1, 2, 3\}$ and multiply each by nine. The time side becomes

$$SG[9A_i^V + B_i^V j l + C_i^V \binom{j}{2} \binom{l}{2} + 9D_i^V \mathbf{1}_{j=l=3} + 54\delta].$$

Writing $p = P_0$, $q = P_1 - P_0$, the jump side becomes $a_m N^2$ times

$$\begin{aligned} 9pC_i + 3qjC_i + (pj + q\binom{j}{2})[-3D_{i-1} + 3lD_{i-1} \\ + 3(D_i - D_{i-1})\binom{l}{2} + 3(D_{i-1} - D_i)\mathbf{1}_{l=3}]. \end{aligned}$$

Their difference is required to be nonnegative. These tests include $v = 0$ and all limiting crossings. No positive lower cutoff on the jump size is introduced.

Independently, nonnegativity of U_t follows by adding 6δ to each spatial cubic Bernstein coefficient of V_i :

$$A_i^V, \quad A_i^V + B_i^V/3, \quad A_i^V + (2B_i^V + C_i^V)/3, \quad A_i^V + B_i^V + C_i^V + D_i^V. \quad (105)$$

The divisions by three are exact for the raw spline coefficients.

B.7 Rigorous sufficient tests for groups of bins

The following tests reduce the number of arithmetic operations. Each is a proved sufficient inequality on a whole group. Failure only triggers subdivision or the detailed polynomial tests; it never permits a numerical substitute.

First consider bins $k = l, \dots, l + D - 1$, possibly intersected with the final partial bin, where D is a power of two. The cost is bounded by the group's final endpoint cost. For $h = (l + v)\Delta$ with $0 \leq v \leq D$, compare $x - zh$ with $(i + 1/2)\Delta - zl\Delta$. Their difference lies in $[-D\Delta, \Delta]$. If

$z \in [(j-1)/l, j/l]$, the reference point lies in cell $i-j$. Let Q_{i-j} be the nonnegative maximum of the curvature on cells $i-j-D, \dots, i-j+1$. Taylor's integral remainder gives

$$\frac{P(x-h) - P(x) + hP_x(x)}{h^2} \leq \frac{1}{2l^2} \sum_{j=1}^l (2l-2j+1)Q_{i-j}. \quad (106)$$

Curvature is linear on each cell, so its maximum is an endpoint maximum. Prefix sums compute the right side exactly. The lower bound on U_t is the minimum of Eq. (105), plus 6δ , divided by $6a_m$. This proves a sufficient integer test. The whole-jump test uses half the nonnegative maximum curvature on the full cells $i-J-1, \dots, i$, and the endpoint cost P_{J+1}/G ; it covers the small bin too. For just $h \leq \Delta$, cells $i-1, i$ suffice.

There is also a useful chord test which does not require convexity. Put

$$i_L = i - l - D, \quad i_R = i - l + 1, \quad W = D + 1.$$

The point $x-h$ lies between the knots with these two indices. If $P_{xx} \geq -m$ there, then

$$P(y) \leq \text{chord}(y) + m(y - x_L)(x_R - y)/2 \leq \text{chord}(y) + m(x_R - x_L)^2/8.$$

Indeed, subtracting the middle expression leaves a convex function with zero endpoint values. Let n_- be the nonnegative opposite of the smallest raw second difference at the intervening knots. Then $m = N^2 n_- / S$, and a sufficient raw correction is

$$C_{\text{chord}} = \lceil 3n_- W^2 / 4 \rceil.$$

Using the raw knot values A_L, A_R , the raw jump remainder is bounded above by $1/W$ times

$$\begin{aligned} & W(A_L + C_{\text{chord}}) + (A_R - A_L)(r - v + D) \\ & - W(A_i + B_i r + C_i r^2 + D_i r^3) + W(l + v)(B_i + 2C_i r + 3D_i r^2). \end{aligned} \quad (107)$$

Set $v = Dz$, $0 \leq z \leq 1$. Compare the group's maximum cost times Eq. (107) with $Wh^2 U_t$. This has bidegree $(3, 2)$ in r, z and is verified by its twelve tensor Bernstein coefficients. Apply this chord test only when the padded indices satisfy $i_L \geq 1$ and $i_R + 2 < n$; otherwise the test fails. If the true remainder is negative the generator inequality is automatic because $U_t \geq 0$; otherwise replacing the cost by its upper bound and the remainder by Eq. (107) is valid.

For an explicit form, let p_j and v_j be the four spatial Bernstein coefficients of the source and time cubics, and let

$$d_j = B_i + 2(C_i/3)j + D_i \binom{j}{2} \quad (0 \leq j \leq 3), \quad \Delta A = A_R - A_L.$$

The coefficients of six times Eq. (107), indexed by $j \in \{0, 1, 2, 3\}$, $k \in \{0, 1, 2\}$, are

$$6W(A_L + C_{\text{chord}} - p_j) + 2\Delta A j + 6\Delta A D - 3\Delta A D k + (6Wl + 3WDk)d_j.$$

The three Bernstein coefficients of $(l + Dz)^2$ are $l^2, l(l + D), (l + D)^2$. These formulas give all twelve exact tests after clearing denominators. For a group truncated by the final cap the tested rectangle can be larger than the required set, which is safe.

B.8 Offset construction and all spatial tails

For each time interval and each of its three temporal components, start $\delta = 0$. Visit spatial cells in increasing order. Enforce Eq. (105), raising δ by an integer ceiling when needed. Apply the whole-jump sufficient test; if it fails, apply the small-jump sufficient test and, when needed, the exact small-jump polynomials. Visit groups starting at 1, 65, 129, ..., each of width 64, intersected with $\{1, \dots, J\}$. First apply the integral test and then, if needed, the chord test. If both fail, bisect the untruncated power-of-two interval and visit its left half before its right half, omitting a half whose start exceeds J . At a single bin use the exact degree-five polynomial tests and raise δ as necessary. Raising δ cannot invalidate an earlier test because its multiplier is nonnegative everywhere, and strictly positive in the coefficient tests requiring division.

After the three components are completed, increase the last δ by at most two so their sum is divisible by three, and integrate Eq. (101). The construction uses only exact integers; no supplied offset array is trusted.

Pad each stored coefficient interval by $J + 7$ exact affine entries on each side. If n is the padded count, the construction visits padded cell indices $J + 3, \dots, n - 3$. In every omitted far-left cell, the complete spline and all allowed downward destinations lie in the zero tail. In every omitted far-right cell, their full stencils lie in the right affine tail. Their remainder is zero and their time derivative is $E' \geq 0$. All joins between stored data and affine tails are checked explicitly. This proves coverage of the entire real line.

Each time interval can be checked independently: its generator inequalities depend only on its three rows and derivative corrections, not on the accumulated constant $E(t_m)$. Summing all increments in increasing time order reconstructs the global offset.

B.9 The objective on every time-interval interior

At the state zero, one coefficient row gives

$$P(0) = \frac{C_{-2} + 23C_{-1} + 23C_0 + C_1}{48S}.$$

Combining Eq. (100) and Eq. (101), the zero-state value on each interval is a cubic $F(\tau) = f_0 + f_1\tau + f_2\tau^2 + f_3\tau^3$, with exact rational coefficients. No initial time shift or subtraction is used.

Let $[u, v] \subset [0, 1]$ correspond to the intersection of this time interval with the desired objective range. Put

$$\epsilon_m = \frac{(v - u)^2}{8} \max\{0, -2f_2 - 6f_3u, -2f_2 - 6f_3v\}. \quad (108)$$

The second derivative is affine, so F exceeds the chord joining its endpoint values by at most ϵ_m . The proof is the chord estimate from the preceding subsection, now applied in time. It therefore suffices to verify the exact rational inequalities

$$(F(u) + \epsilon_m)^2 \leq c^2 t(u), \quad (F(v) + \epsilon_m)^2 \leq c^2 t(v), \quad (109)$$

with nonnegative augmented values. The augmented chord majorizes F ; concavity of $\sqrt{t}$ places it below $c\sqrt{t}$. This proves the objective at every intermediate time, including a partial first or last interval.

B.10 The resulting bounds

The recurrences above produce 4618 fine knots and 2166 coarse knots. Every scalar test and elimination-divisor check holds. The three temporal components pass the spatial and jump in-

equalities on all 4617 fine slabs. Of these, 1380 intersect $[1/8, 499/1000]$, giving 2760 augmented endpoint tests. Their largest squared ratio is

$$\frac{108413955008912677277181}{847461363712000000000000} < \left(\frac{357671}{1000000}\right)^2 = c_A^2.$$

The maximizing time, augmented value, and final integrated offset are, respectively,

$$\frac{351936469647}{10000000000000}, \quad \frac{13579843057641}{64000000000000}, \quad \frac{352096853}{10000000000000}.$$

For the unchanged coarse construction, the 3000 augmented endpoint tests on $[499/1000, 30]$ give a ratio strictly below $0.355195272716 < 89/250$. Thus the two closed objective ranges meet without a gap.

The initial coefficient rows and the nonnegative spline weights give $U(0, x) \geq x_+$. The finite offset rules establish $U_t \geq 0$ and Eq. (84) on every state and jump cell, including the affine tails. Temporal Bernstein positivity and the augmented chord tests extend these inequalities and objectives throughout every time interval. This proves Lemma 4.28. Every scalar input, coefficient, and offset is determined by the rational rules printed here; verification is a finite arithmetic calculation requiring no external data.

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